



UNIVERSIDAD CARLOS III DE MADRID
ESCUELA POLITÉCNICA SUPERIOR

GRAPHICAL EXPRESSION

FAN ASSEMBLY

AUTHORS

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(12/12/2023)

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For the development of this project, we selected a house fan to be designed in Solid Edge (the modeling software). For the design, we meticulously created 3D models for each part, ensuring that the sizes and shapes were precise. Subsequently, with the Solid Edge *draft*, we generated detailed drawings for each part using templates created for proper labeling. We assembled all these parts using *assembly* to ensure seamless functionality. Following that, we utilized *draft* again to produce drawings for the entire fan, encompassing exploded views with labels and lists.

Throughout the project, we acquired skills in 3D modeling, assembly simulation, and drafting as we familiarized ourselves with the Solid Edge environment.

DESCRIPTION OF EACH PART

Foot1

It is joined to foot2 to form the base of our object in order to allow its stability. It is also joined to the “foot_stick through” 2 screws (“foot_screw”), and to 2 “foot_detail”.



Foot2

When it's joined to “foot1”, it forms the base of our object in order to allow its stability. It is also joined to the “foot_stick” through 2 screws (“foot_screw”), and to 2 “foot_detail”.



Foot_detail

This piece is connected to both feet (“foot1”, “foot2”) by its extremes, and they are used to beautify the aesthetics of the base. (4 units).



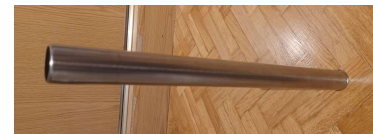
Foot_stick

It's the piece in charge of the connection of the base with the controller part. It is connected to the feet (“foot1”, “foot2”) through 4 screws (“foot_screw”), and hides the “metal_stick” and the “gear_stick”.



Metal_stick

It makes possible the manual height regulation of our object. It is inside the “foot_stick”, and thanks to the “gear_foot”, it has a limited height. Finally it is connected to “central handler”.



Gear_stick

This piece surrounds the “metal_stick”, and it is used to limit the height, thanks to this piece, the “metal_stick” doesn't separate from the “foot_stick”.



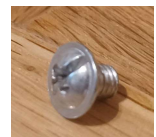
Connector

It is used in order to help in the manual regulation of the height. The tighter you screw this piece to the “foot_stick”, the better it will hold the main part of the fan at the desired height.



Foot_screw

Screw that joins the “foot_stick” to the base composed by “foot1” and “foot2”. (4 units).



Screw central node

This screw with the nut and the "cap" performs as an union for the upper part (fan, motor) and the controller (4 button system block).



Central nut

It is used with “screw central node”.



Central cap

Also used with “screw central node”.

"Umbrella"

Screw with a plastic piece attached to its upper part that resembles an umbrella, used with a nut to attach the mid handler to the base of the fan.



"Umbrella" square nut

It's used with “umbrella”.



Central handler front(button).

It is the front piece of an union with the back part that encapsulates the 4 button system and the electronic pieces needed for the correct working of the fan.



Central handler back

It's the back part of the handler and the umbrella is situated on the lower side.



4 button system

While fully operational this system is used to start/stop, and regulate the velocity of the fan, it is attached to the “handler front part”, with 12 mm long screws.



Screw (12mm)

Screw with a star head that unites the “button system” and provides stabilization.



Back lid

The back lid is the first part of the fan that you see, and it is a detachable cover that surrounds the upper section of the fan. Its functions include ensuring safety, enhancing visual appeal, and providing practicality as it includes ventilation grills to allow air circulation.



Front lid

We named the front lid as such because it is the one facing and connecting the motor to the entire helix system. Together with the “back lid”, they form the complete enclosure for the fan's internal component. This part plays a crucial role in facilitating the connection to the helix system, while the back lid complements this arrangement to create a cohesive unit.



Connector one

This connector effectively links the "motor top" and the "front lid" by securely threading them together.



Connector two

The function of this connector is to adjust the tension of the helix, either loosening or tightening it for smoother movement.



Helix

The helix is the part of the fan designed to optimize airflow through the optimal shape of its blades. In our case, it is made up of 6 blades with an inclination of 30 degrees from the center axis. It is joined to the motor by a quite large screw and is one of the main components of the fan.



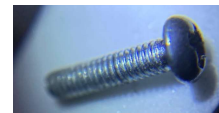
Nut assembly lid

This nut together with the screw join the lids. (6 units)



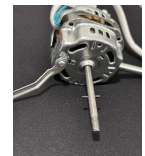
Screw assembly lid

This screw is used to join both the top and the bottom one of the fan. (6 units)



Screw assembly helix

This screw is quite large as it is used to connect the motor with the helix and the "connector two"; therefore, it has to go through the upper part of the fan.



Button_move_fan

Button that controls if the fan moves in a lateral way or if it is maintained static. It is joined to the motor with a screw ("screw_button_move_fan") going through the hole in the upper part of the piece "motor_cover".



Piece_move_fan

Piece of the fan that regulates the up-and-down movement of the fan. It is attached to the motor with a screw ("screw_piece_move_fan") in the lower part of the piece "motor_cover".



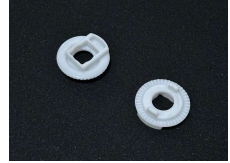
Motor_cover

Cover that protects the motor of the machine and it is attached to it by a screw ("screw_motor_cover"). It has the "button_move_fan" joined through the hole, which is in its upper part and in its lower part it has the "piece_move_fan joined".



Round_thread

Two pieces attached to the laterals of “piece_move_fan and where a screw goes through so these two pieces and “central_switches_cable_cover” are joined.

**Motor_top**

Piece of the fan which covers the frontal part of the motor. It is joined to it by four screws (“screw_motor_top”) and attached to the “motor_cover”. In this piece, the connector 1 is coiled.

**Top_screw_piece_move_fan**

Piece that covers the “screw_piece_move_fan” located in “piece_move_fan”.

**Screw_motor_cover**

Screw that joins the piece “motor_cover” to the motor of the fan (1 unit).

**Screw_piece_move_fan**

Screw that joins the piece “piece_move_fan” to the motor (1 unit).

**Screw_motor_top**

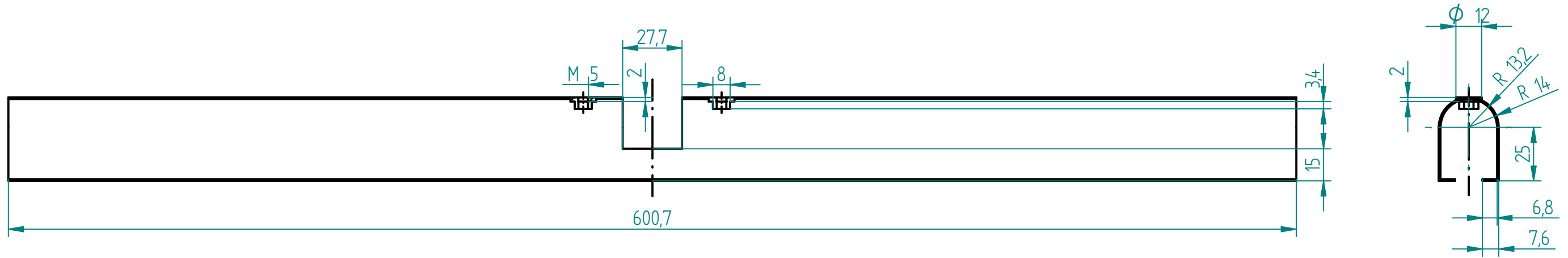
Screw that joins the piece “motor_top” to the motor of the fan (4 units).

**Screw_button_move_fan**

Screw that joins the piece “button_move_fan” to the motor of the fan going through the “motor_cover” (1 unit).



DRAFTS + SUBASSEMBLY
(BASE)

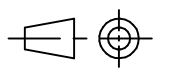


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Foot1

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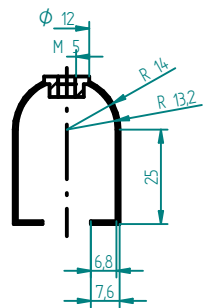
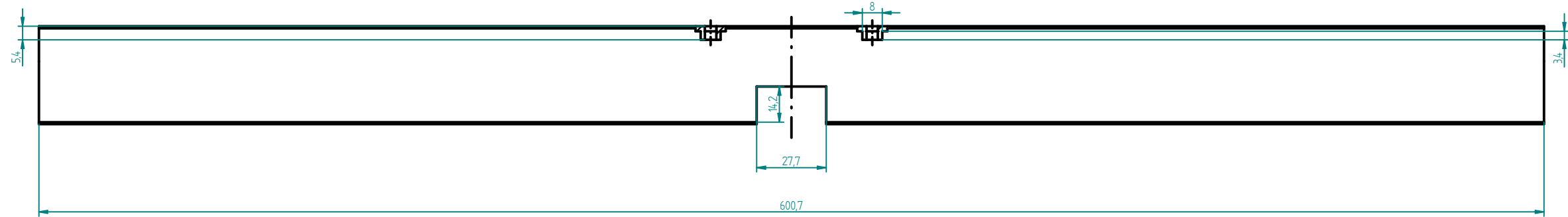
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Kind of draft:

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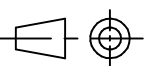


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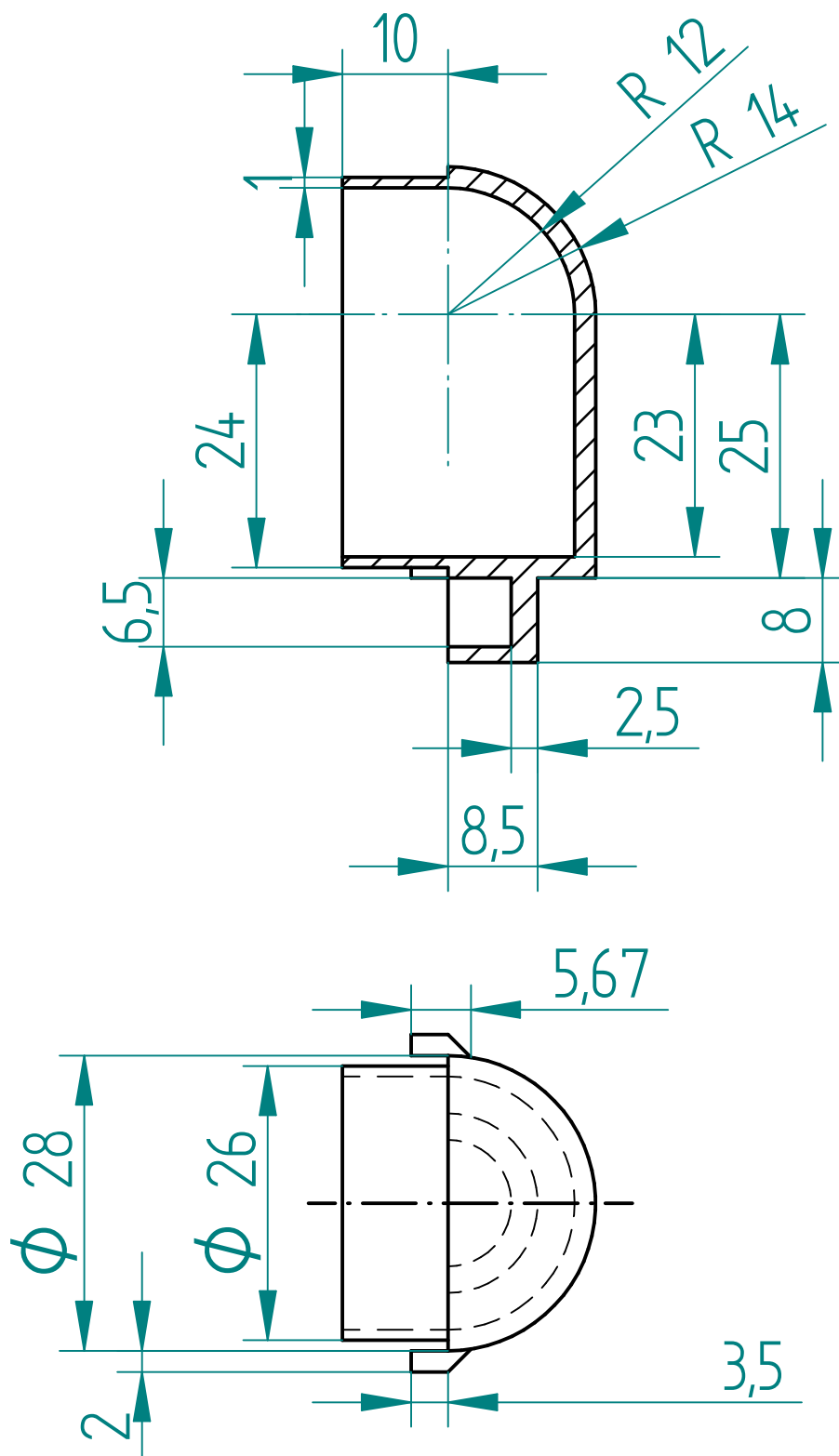
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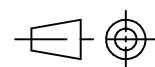
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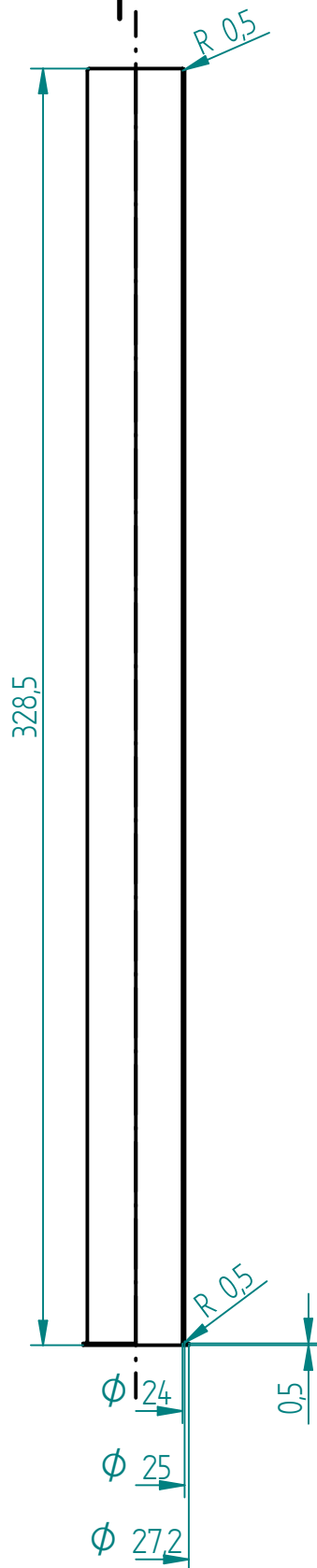
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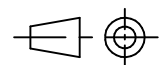


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Title:

Metal_stick



Complementary title:

Scale: 1:1,81

Created by:

Andrea García

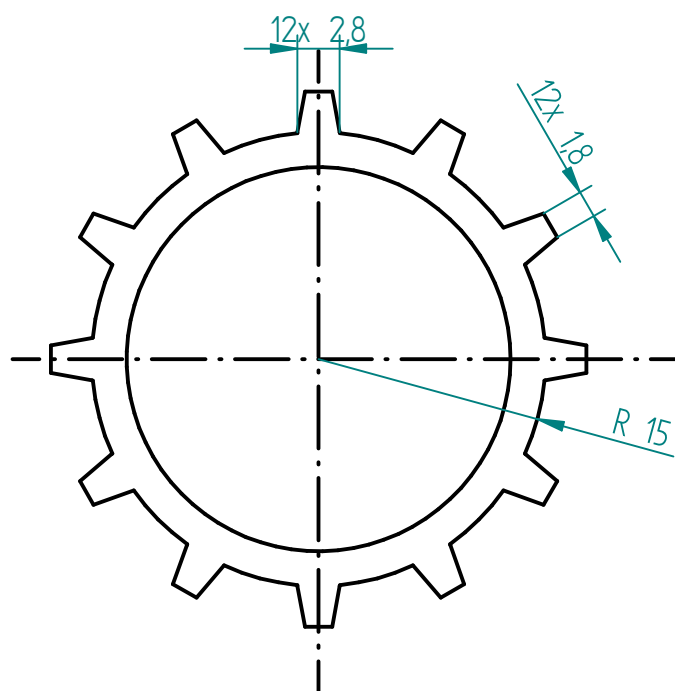
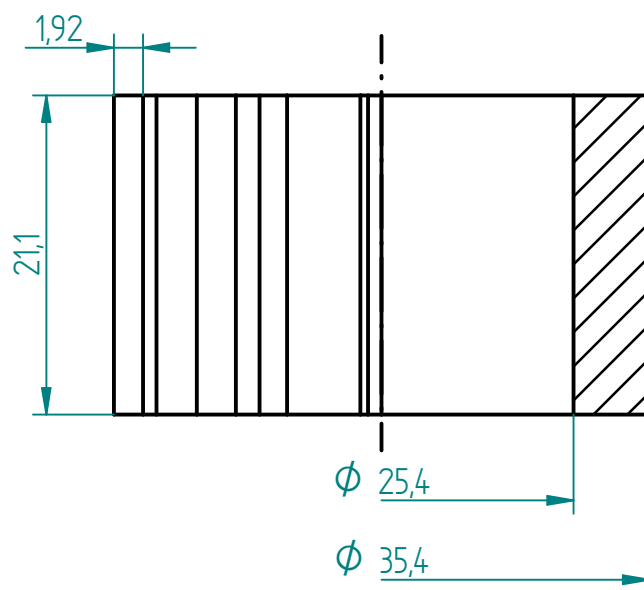
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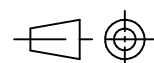
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Gear_stick

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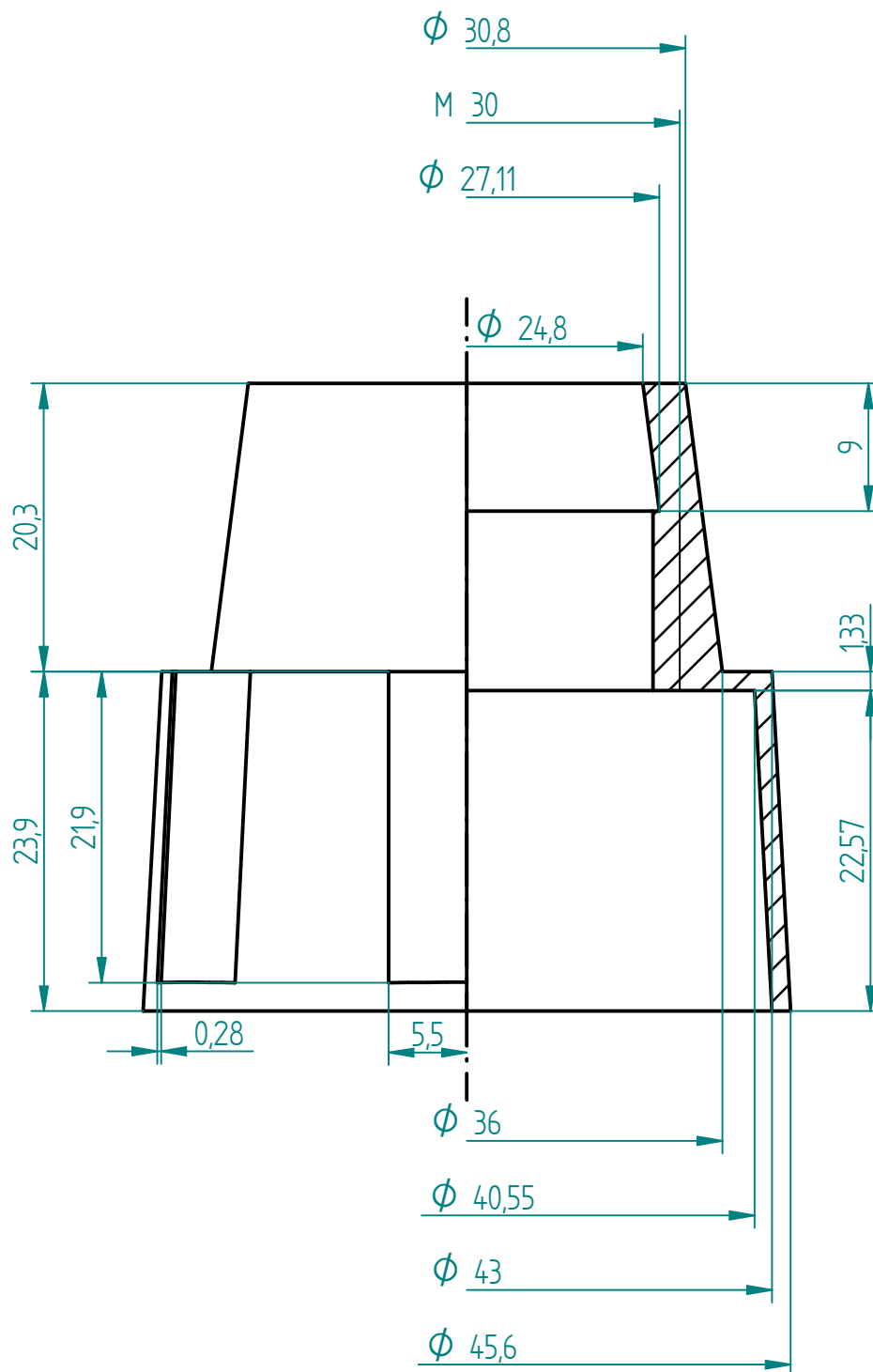
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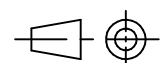
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Title:

Connector

Complementary title:



Scale:

2:1

Created by:

Andrea García

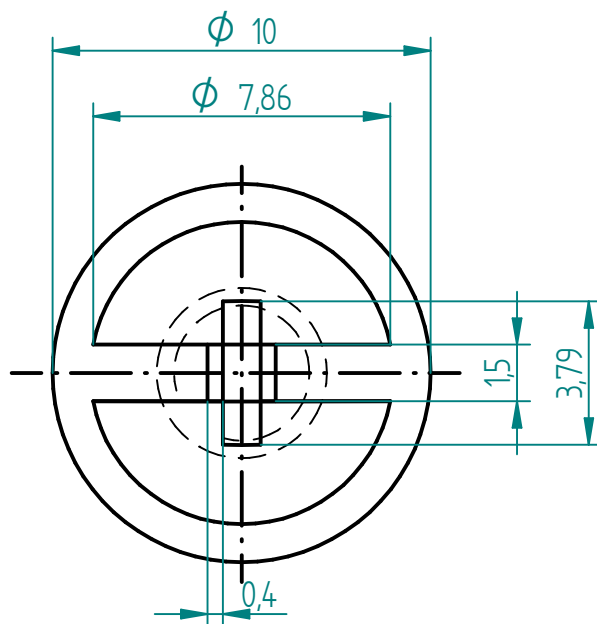
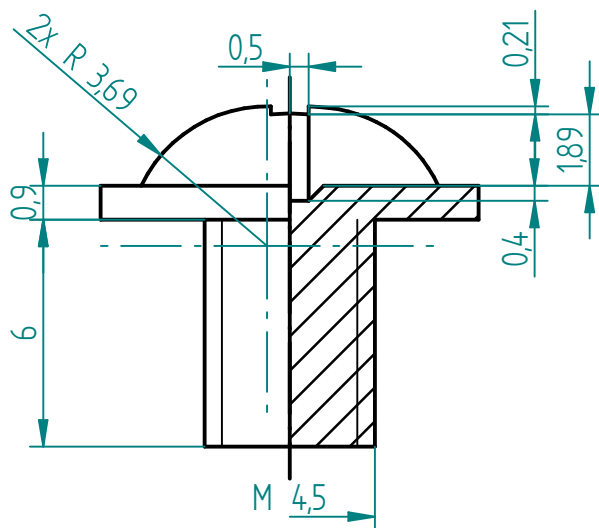
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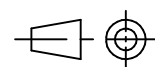
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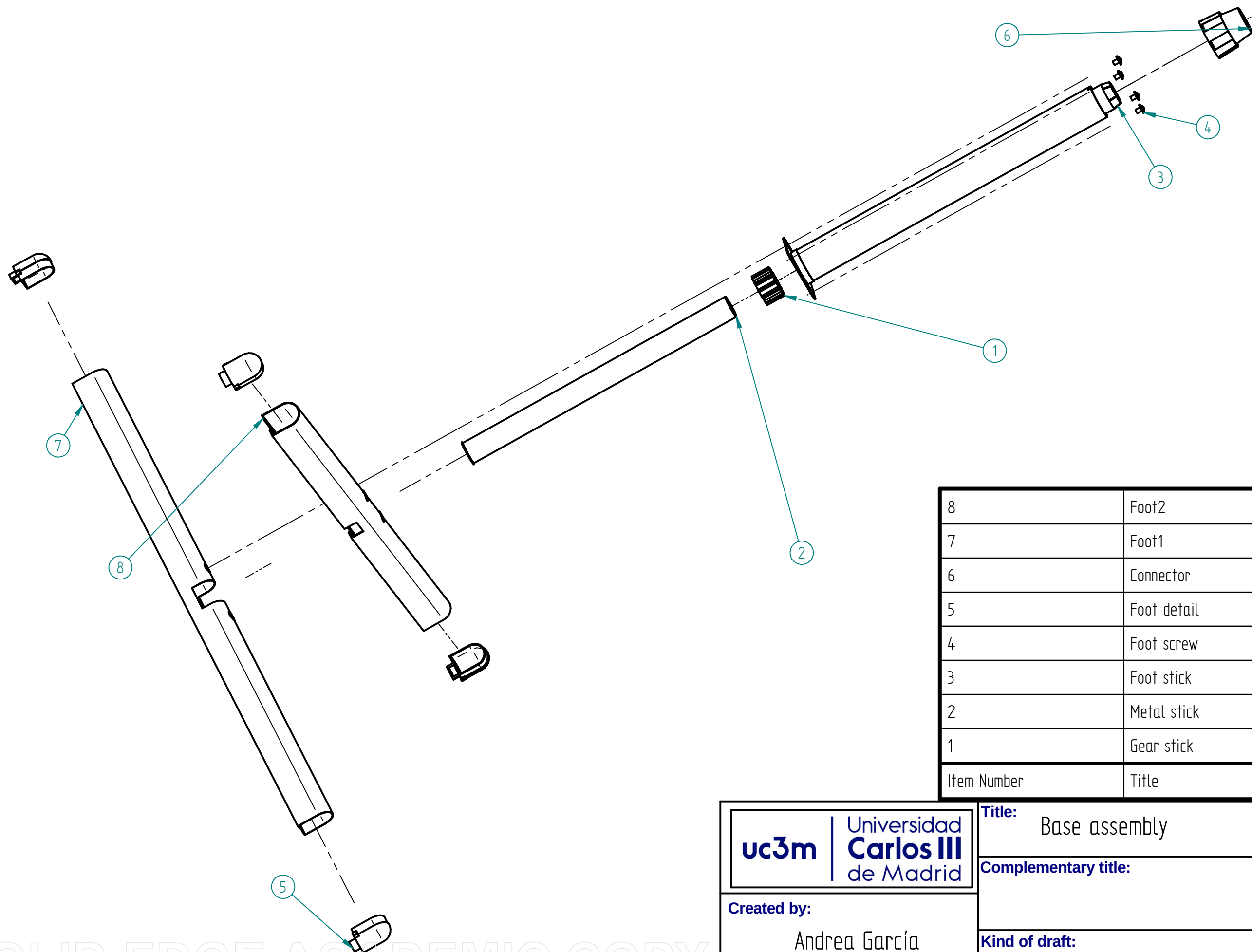
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8	Foot2	1
7	Foot1	1
6	Connector	1
5	Foot detail	4
4	Foot screw	4
3	Foot stick	1
2	Metal stick	1
1	Gear stick	1
Item Number		Title
		Cantidad



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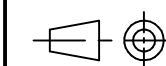
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Base assembly

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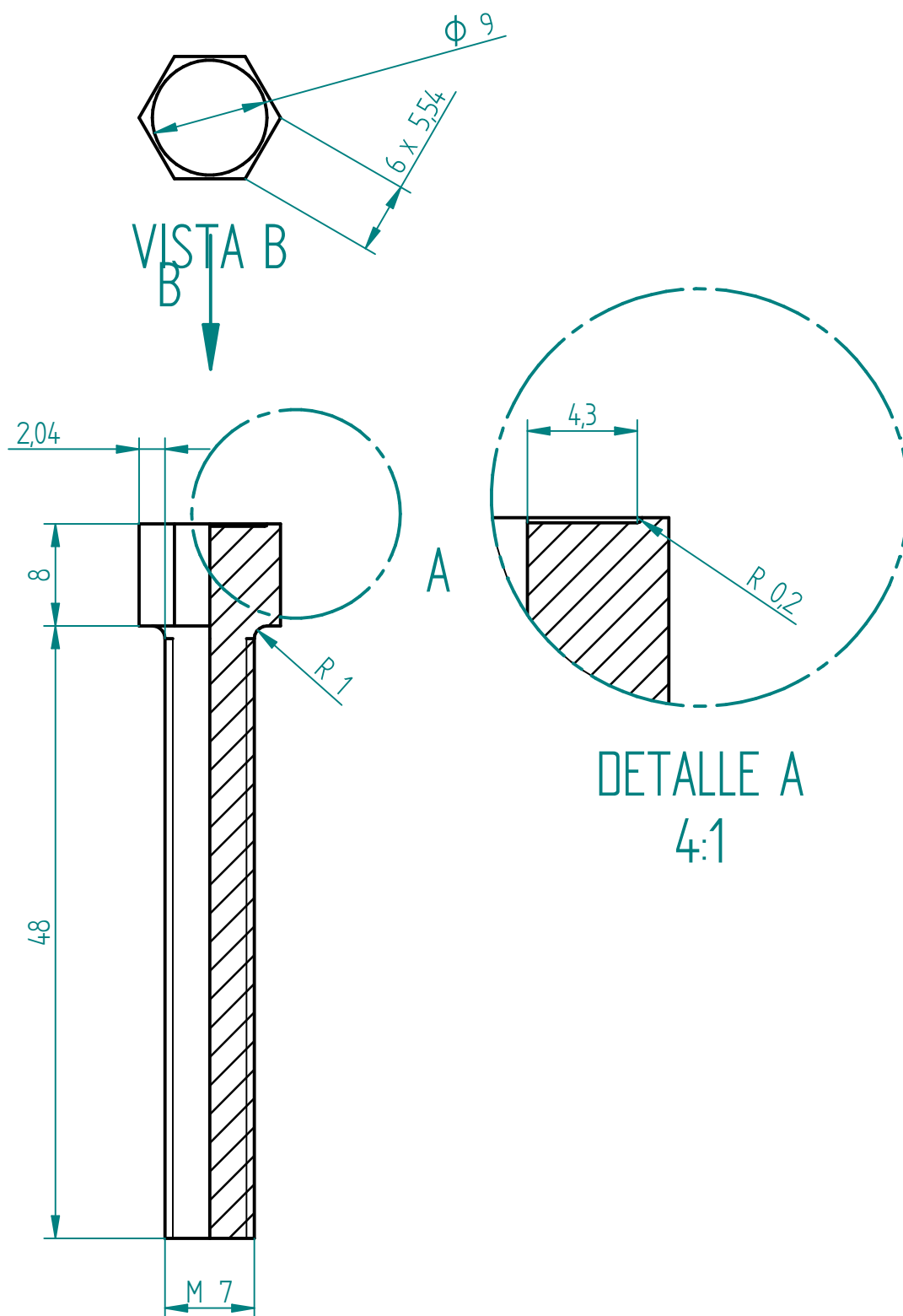


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DRAFTS + SUBASSEMBLY
(CENTRAL HANDLER)



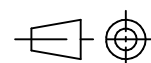
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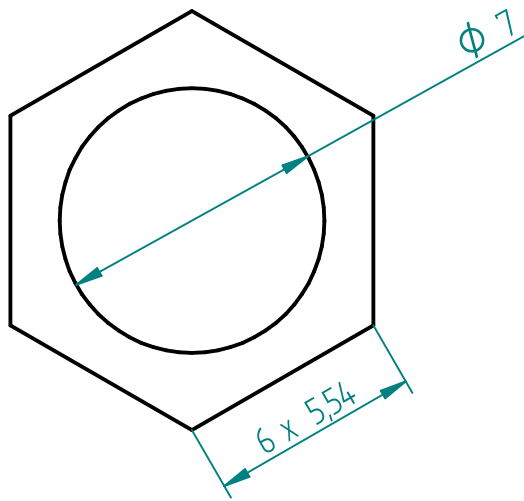
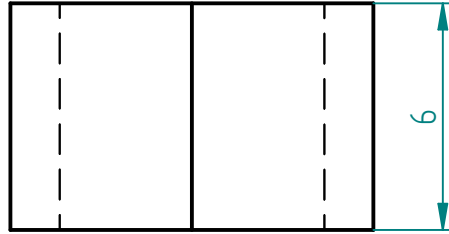
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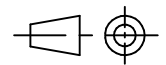


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Complementary title:

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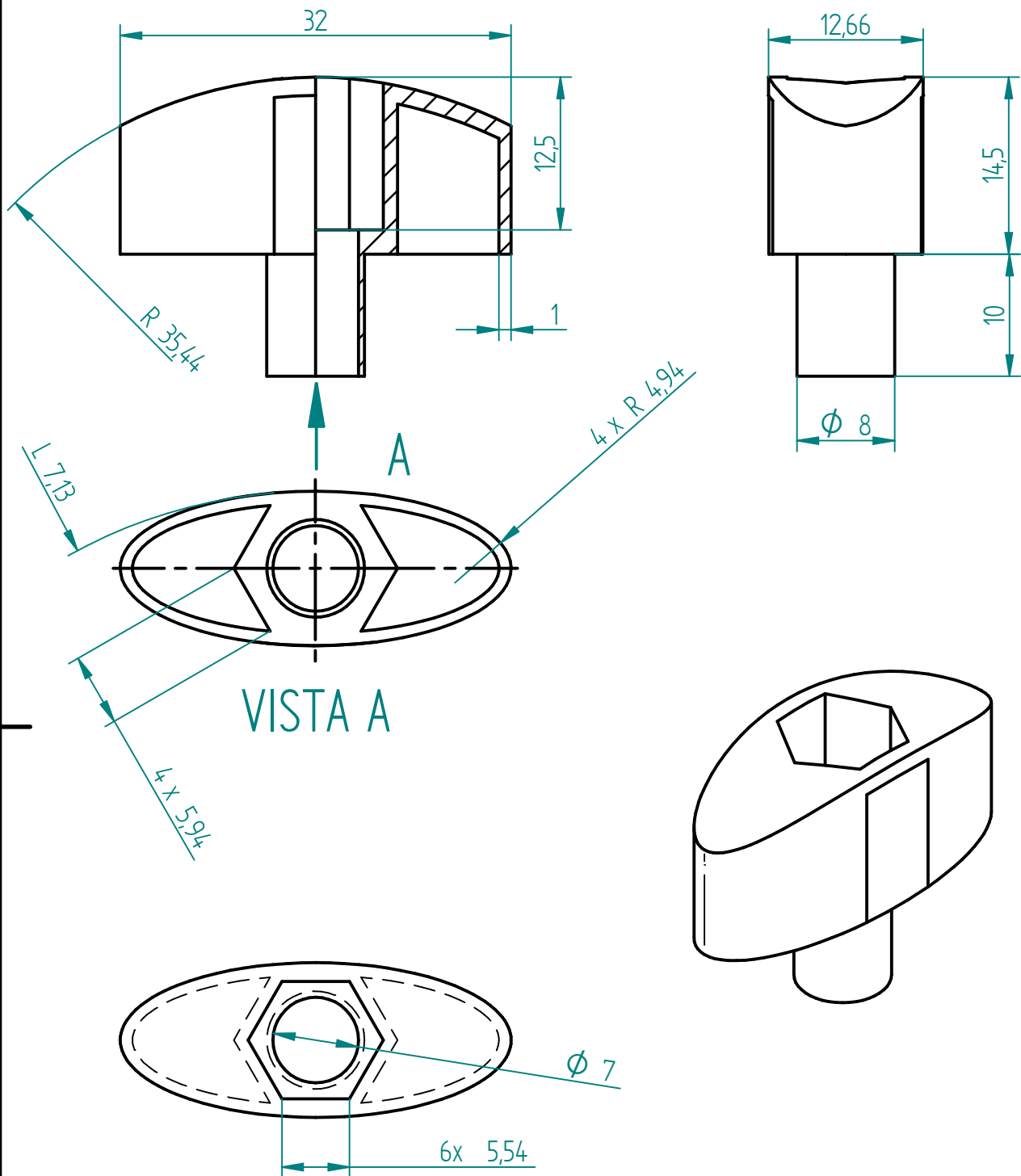
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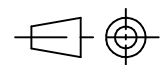
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Complementary title:



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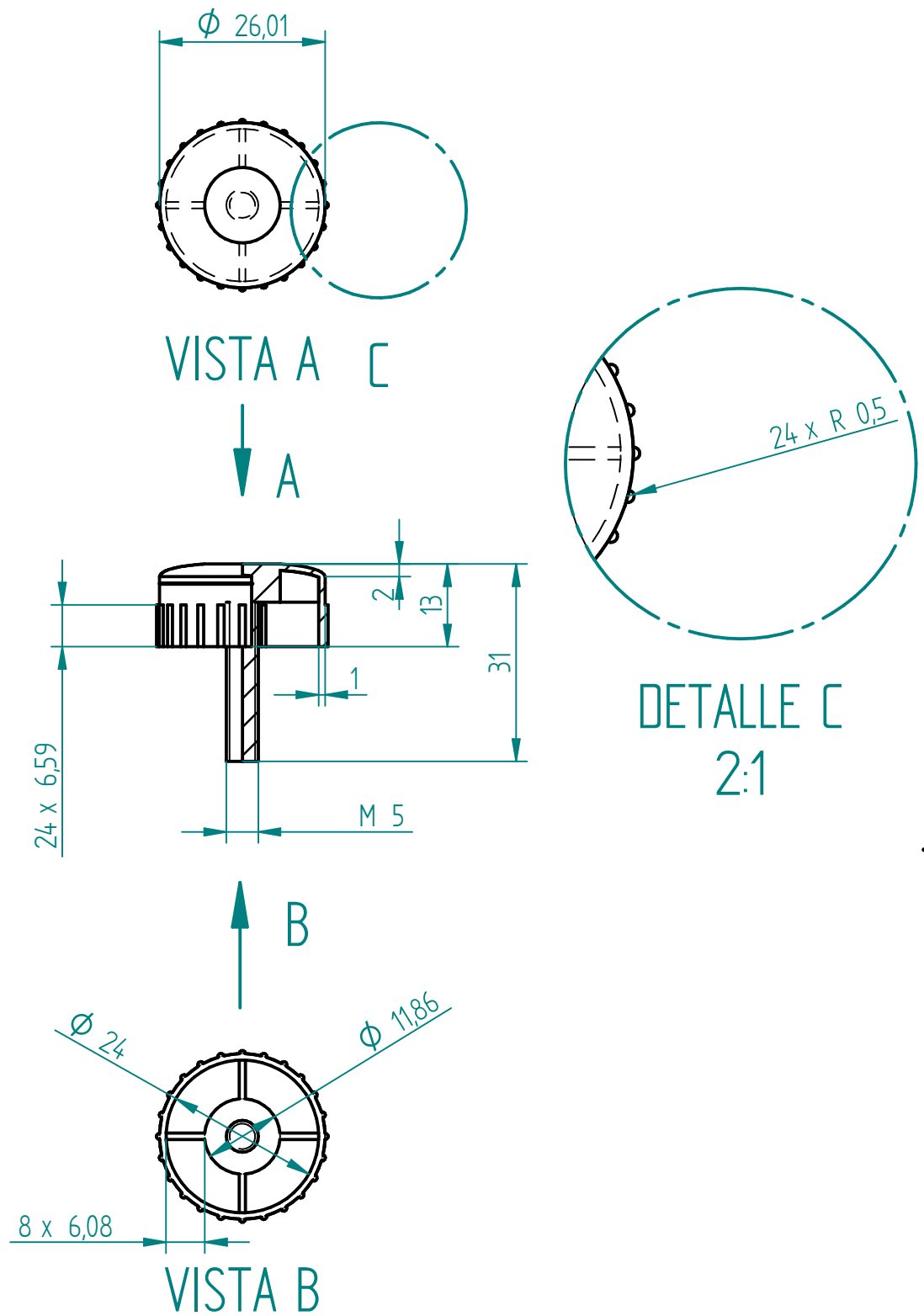
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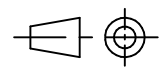
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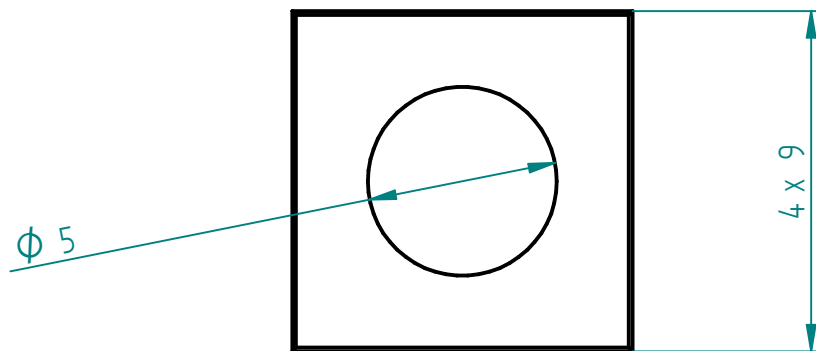
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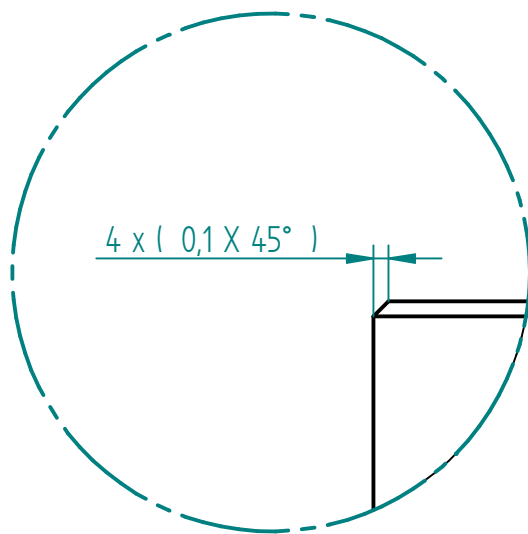
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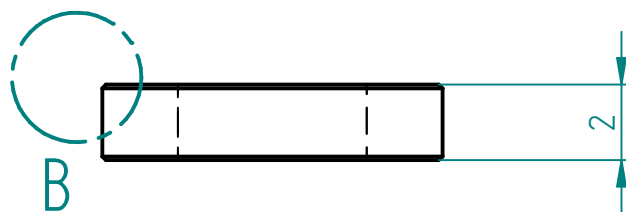
VISTA A



DETALLE B
20:1



A



B

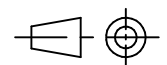
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Title:

umbrella_square_nut

Complementary title:



Scale:

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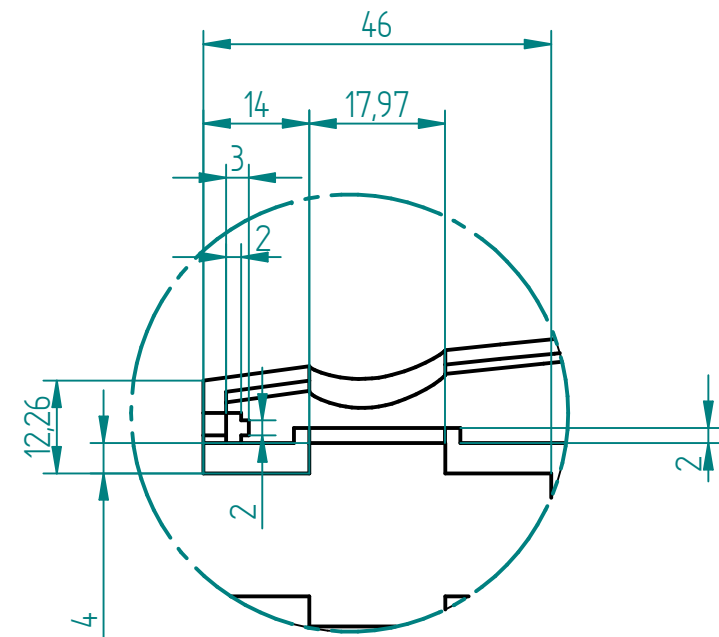
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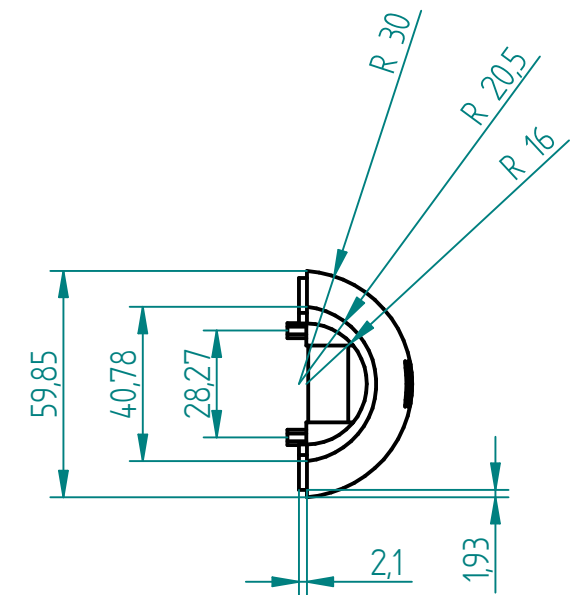
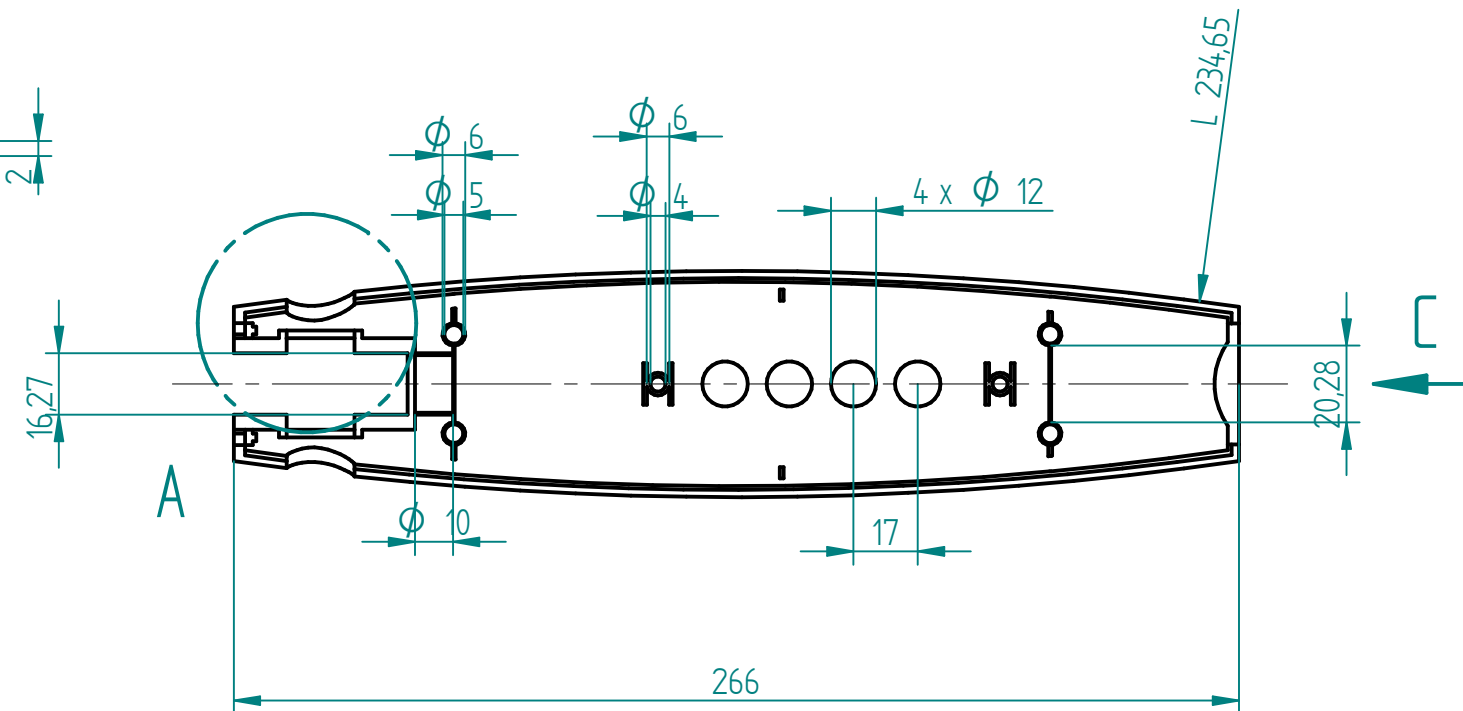
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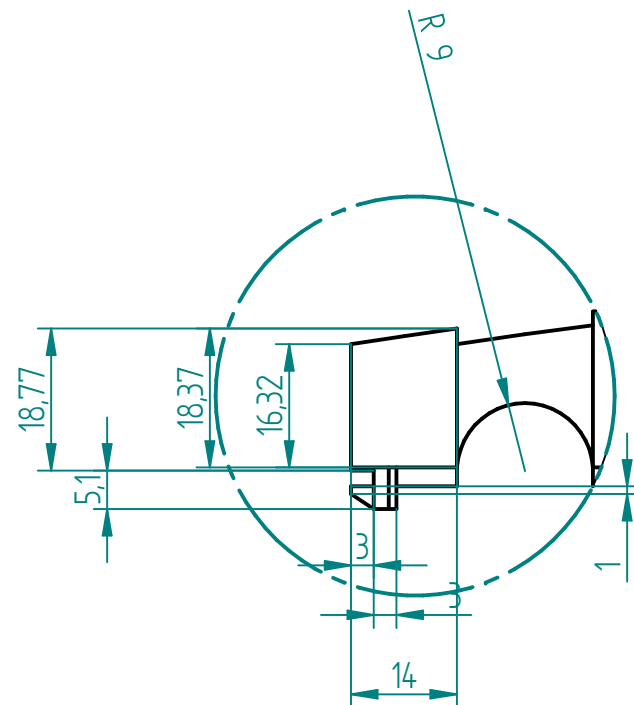
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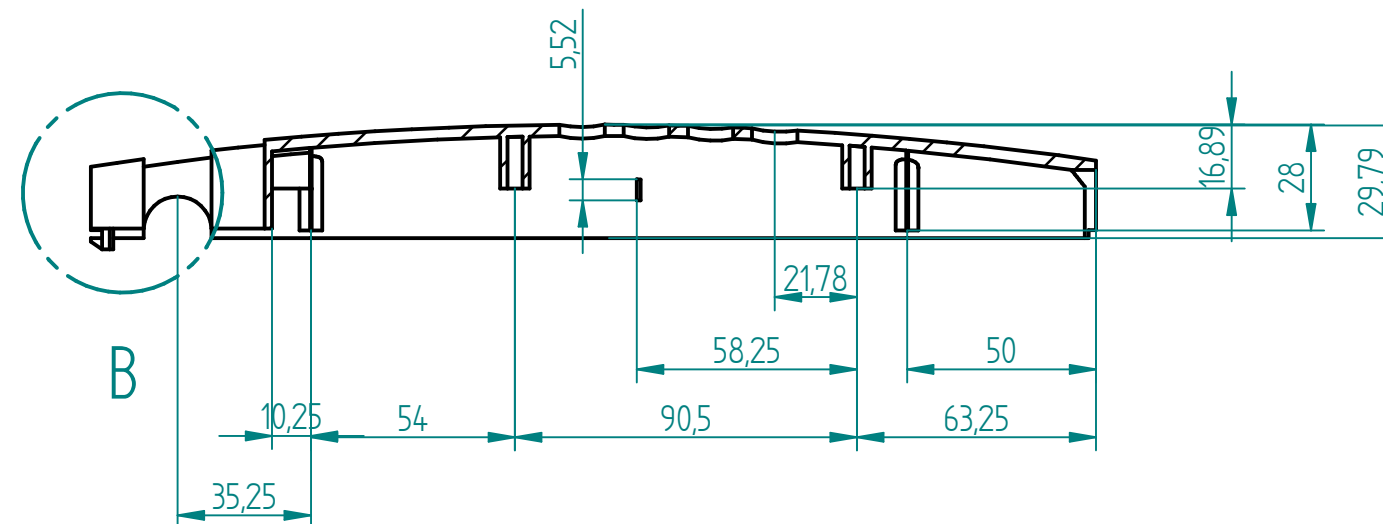
DETALLE A
1:1

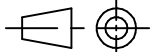


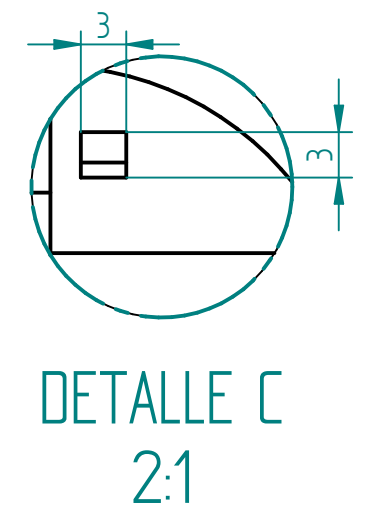
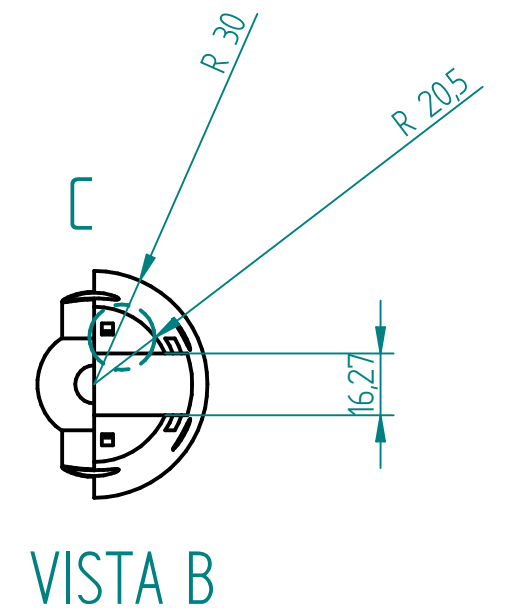
VISTA C



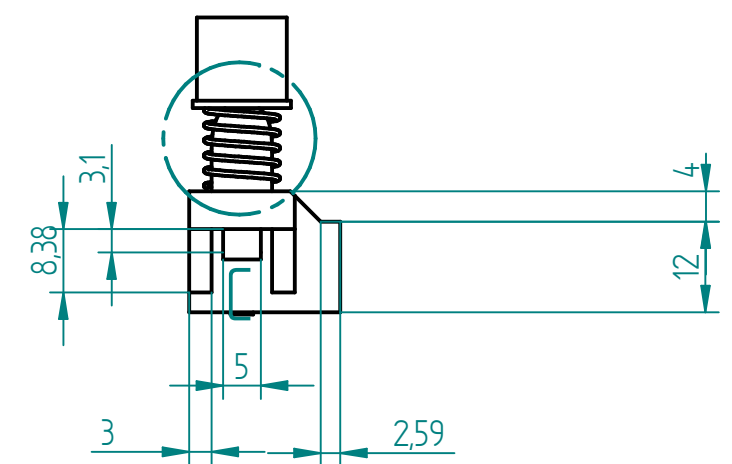
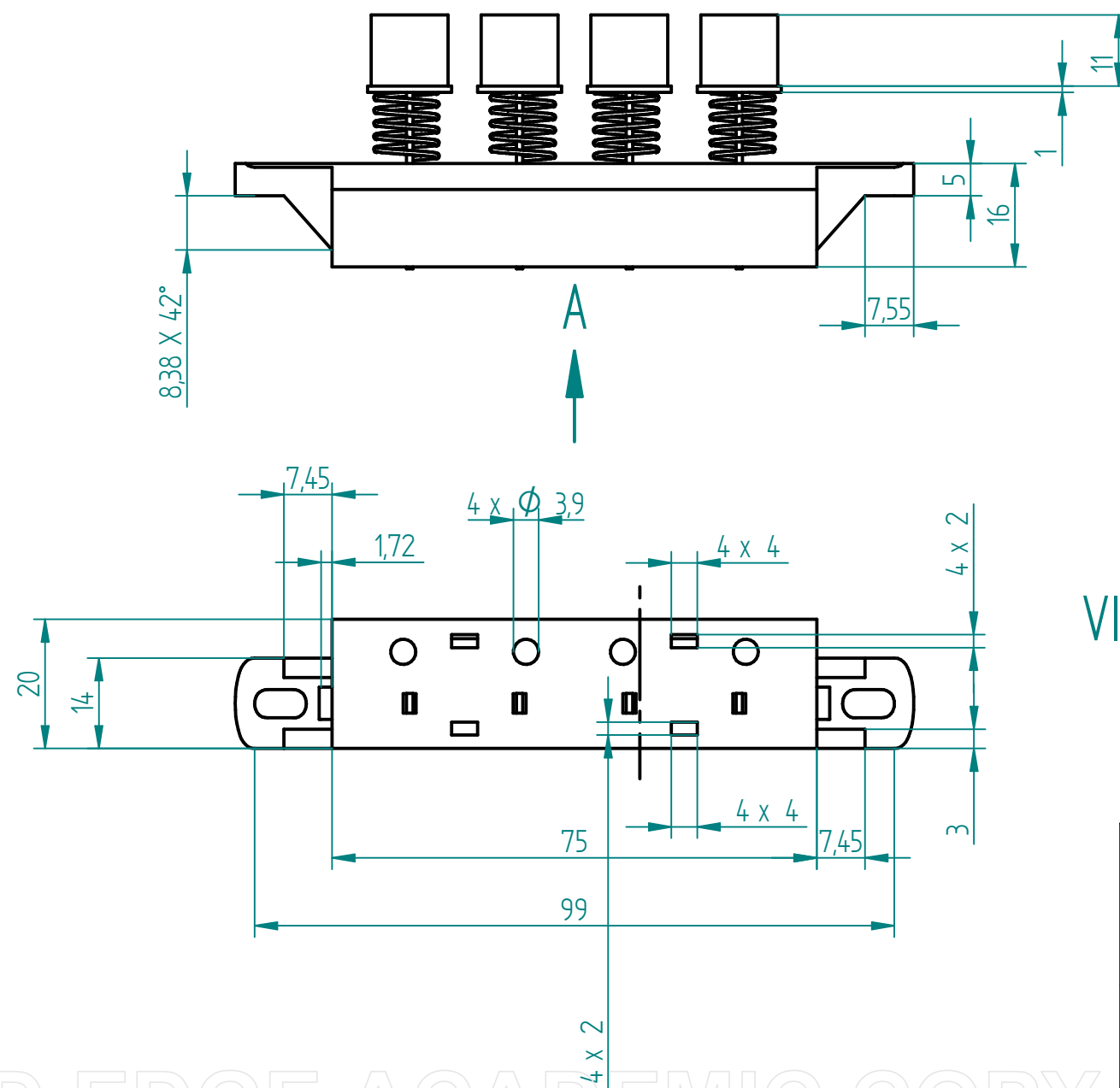
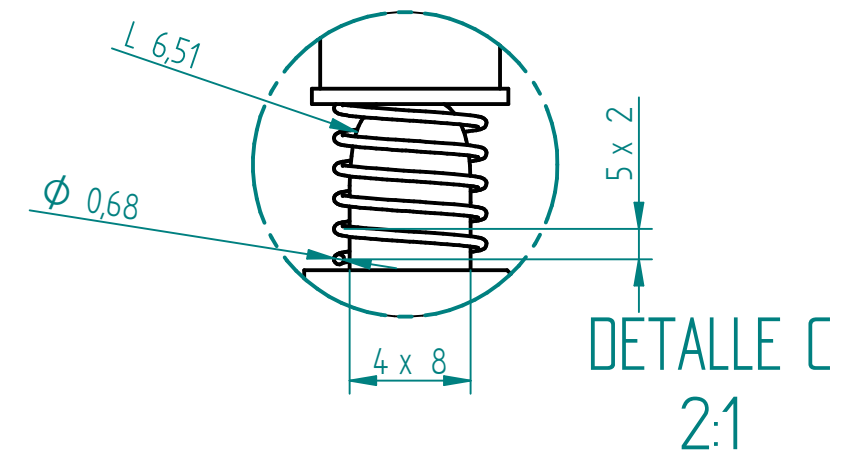
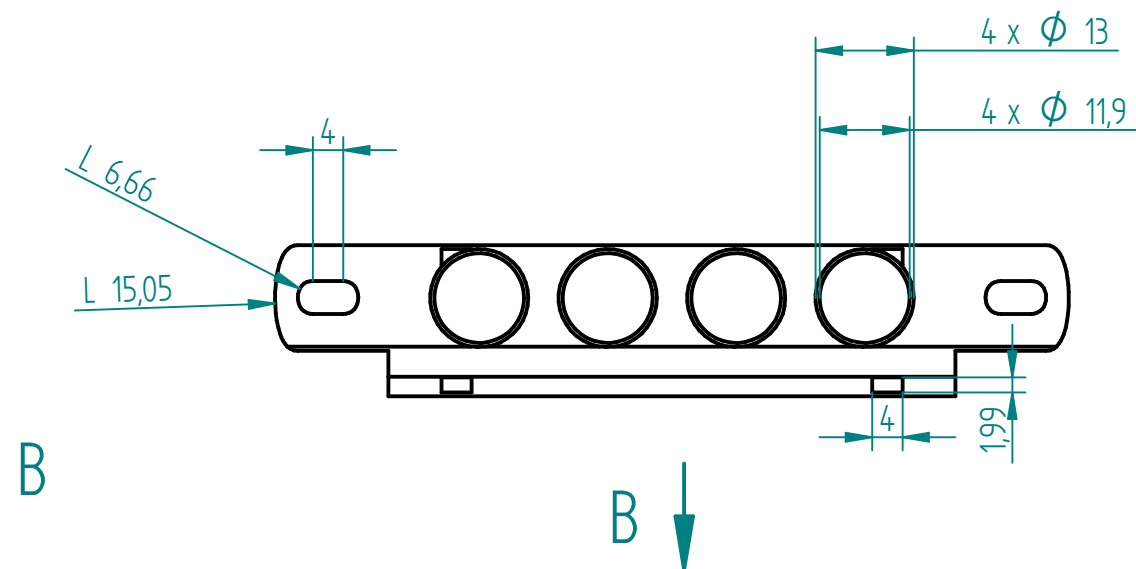
DETALLE B
1:1



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Created by: Jin weijian victor chen	Kind of draft:		Scale: 1:2
			Date: 08/12/2023
			Page number: 25



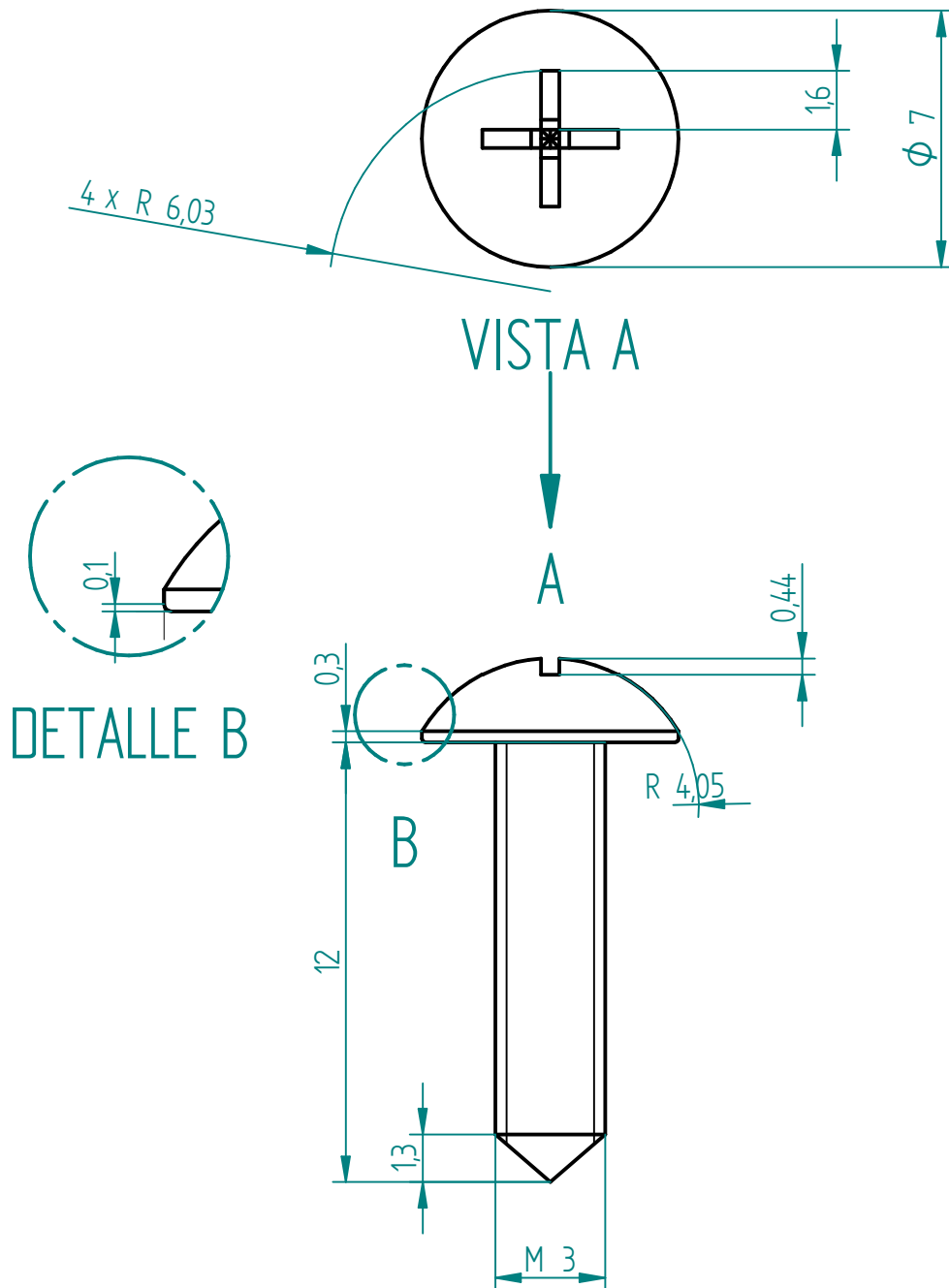
VISTA B



VISTA A

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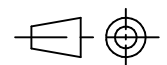
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Scale:

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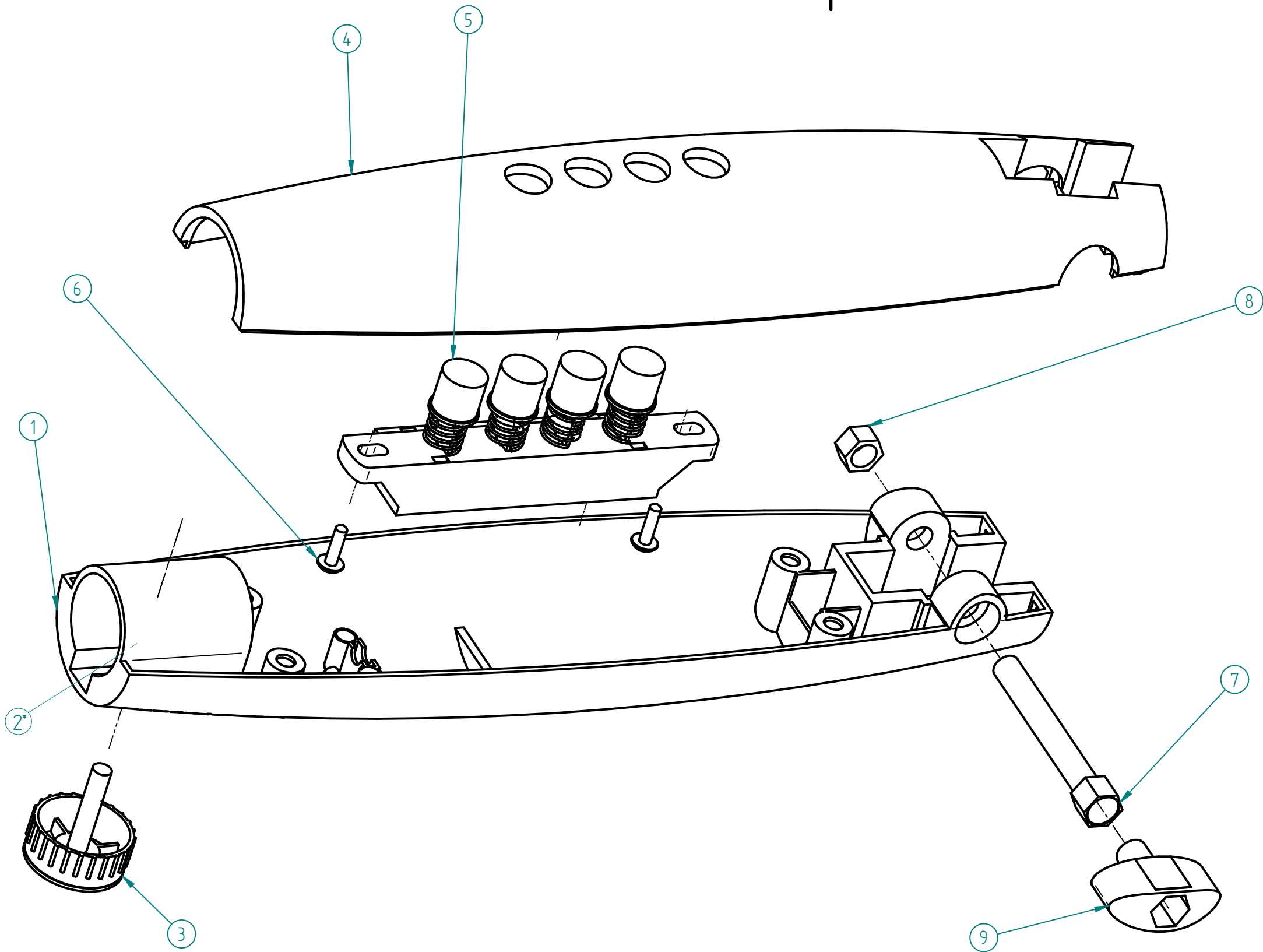
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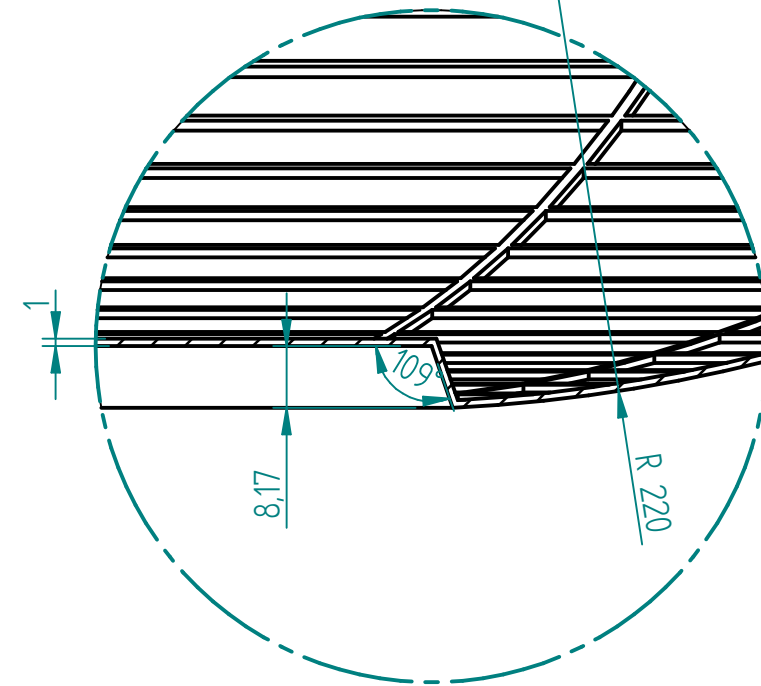
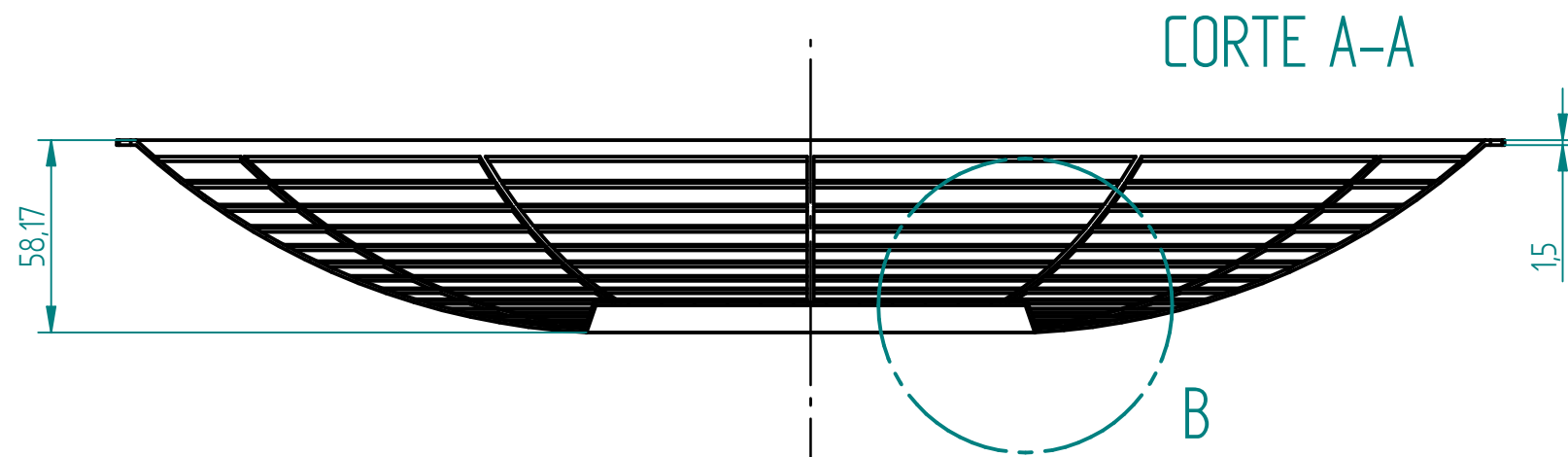
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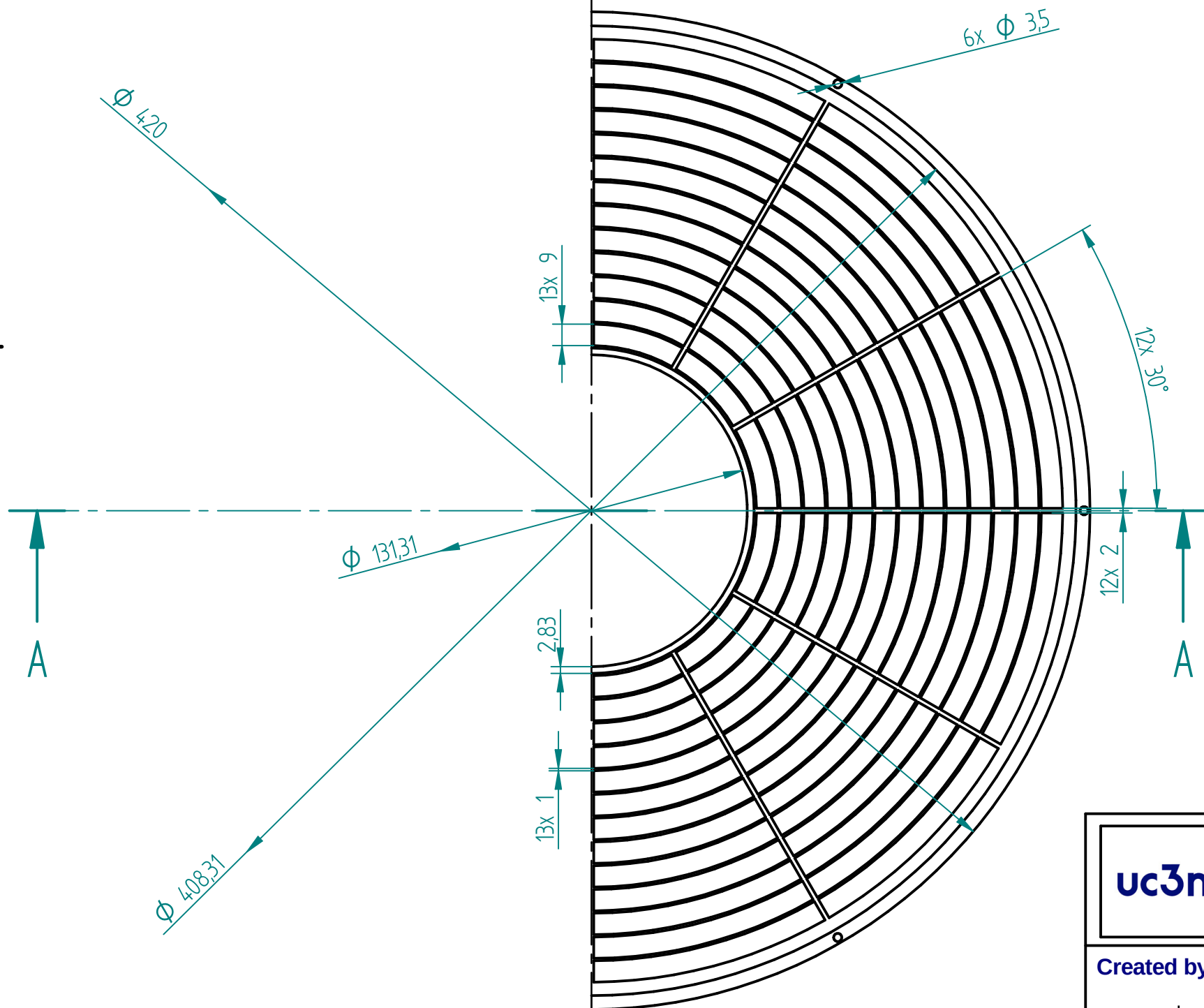


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8	central_nut	1
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6	screw_(12mm)	2
5	4_button_system	1
4	central_front(button)	1
3	umbrella	1
2*	umbrella_square_nut	1
1	central_back	1
Número de elemento	Título	Cantidad

DRAFTS + SUBASSEMBLY
(HELIX AND MOTOR)



DETALLE B
1:1



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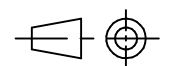
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Lola Martínez

Title:

BACK LID

Complementary title:



Scale:

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Date:

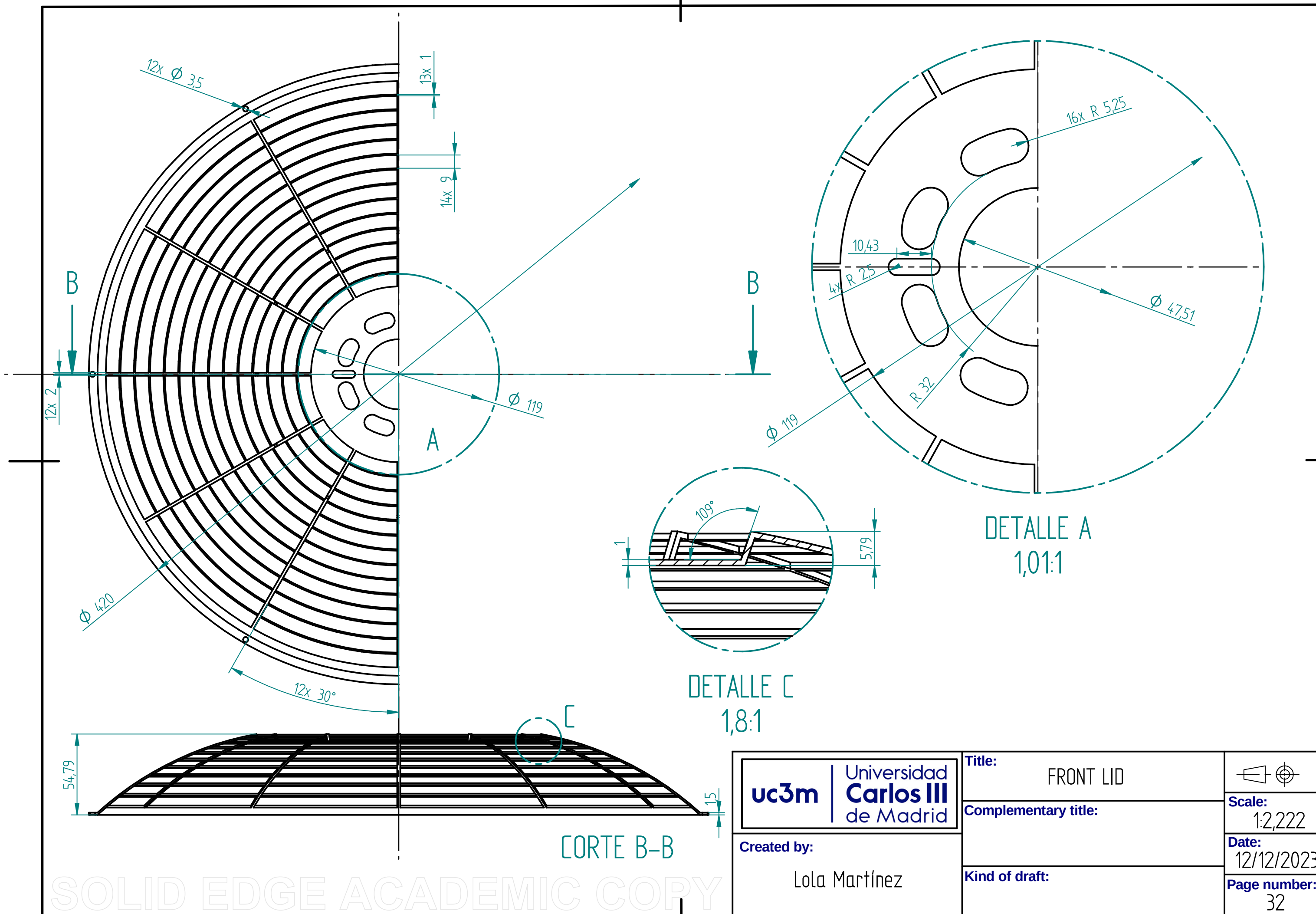
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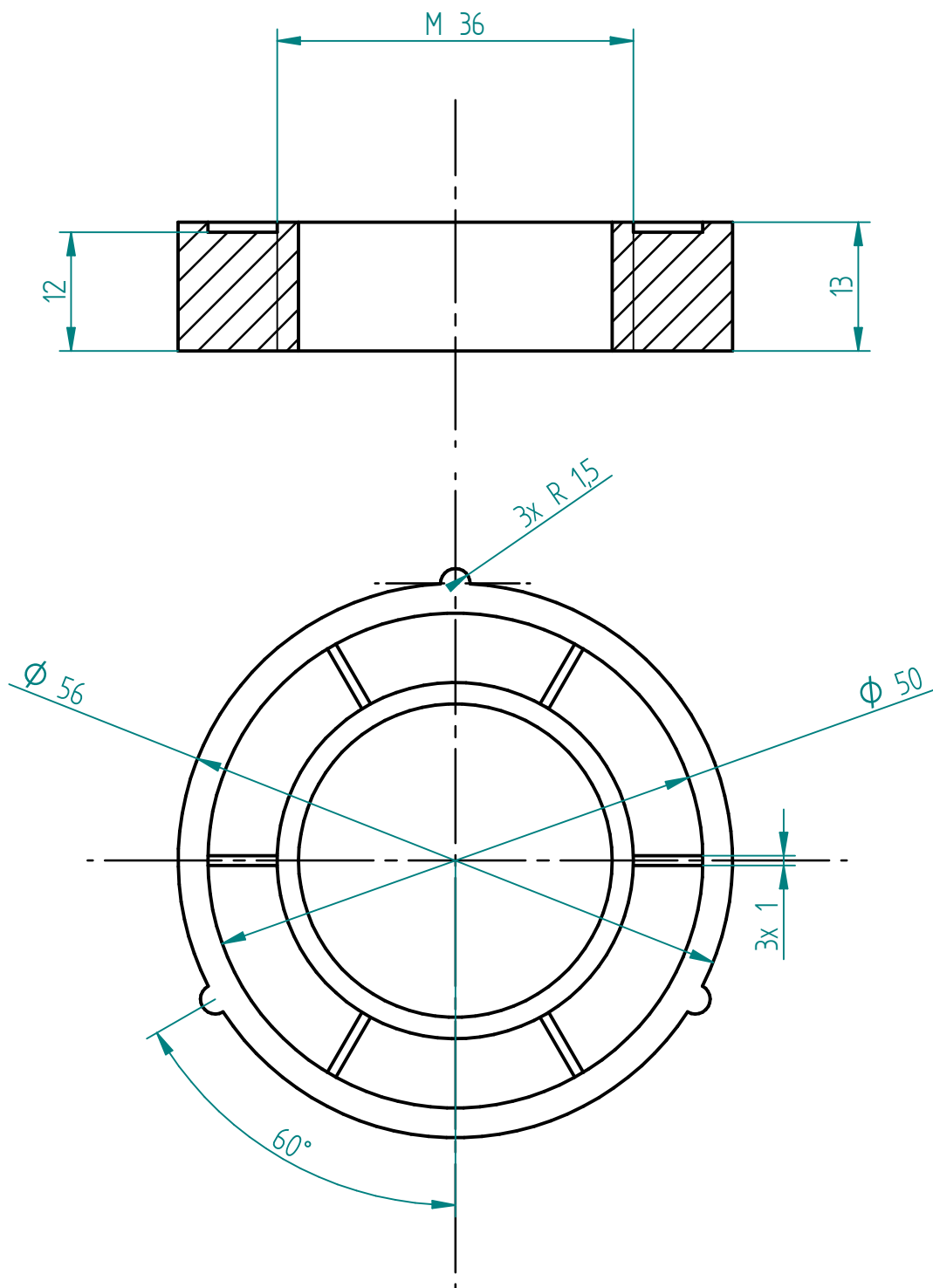
31

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SOLID EDGE ACADEMIC COPY

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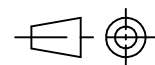
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Complementary title:



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Created by:

Lola Martínez

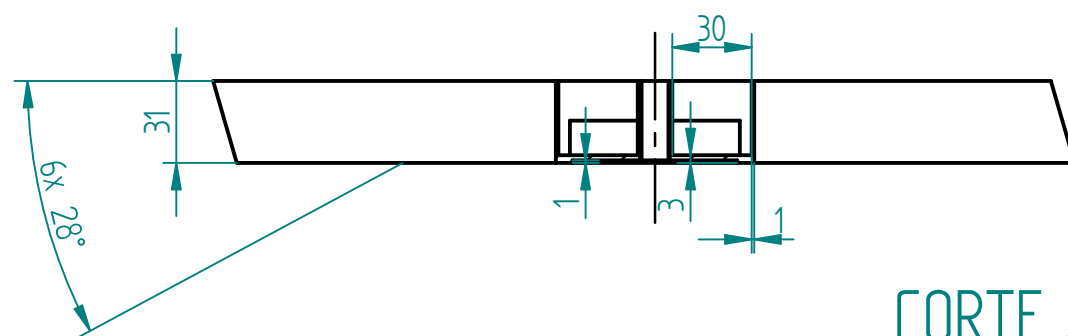
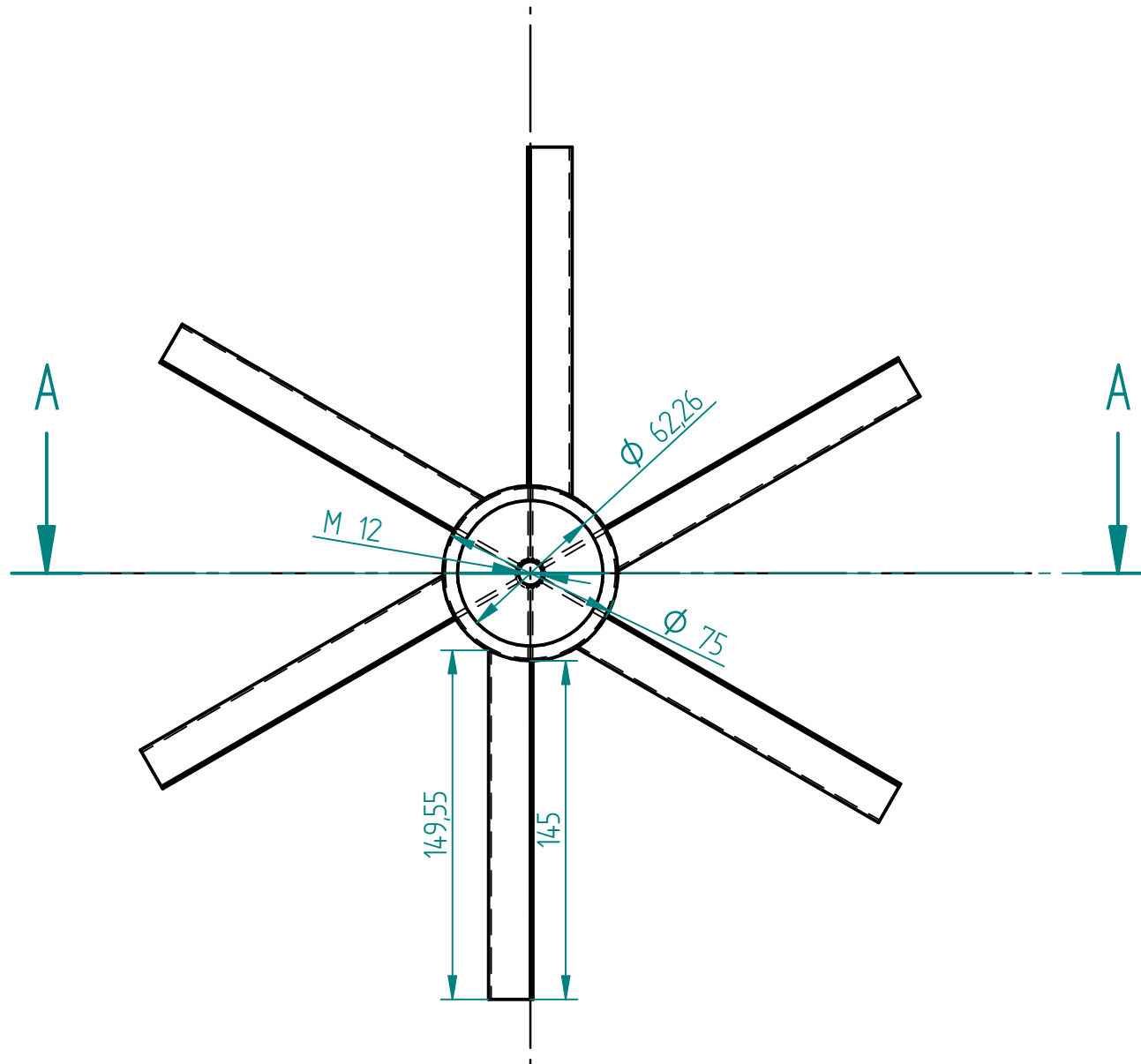
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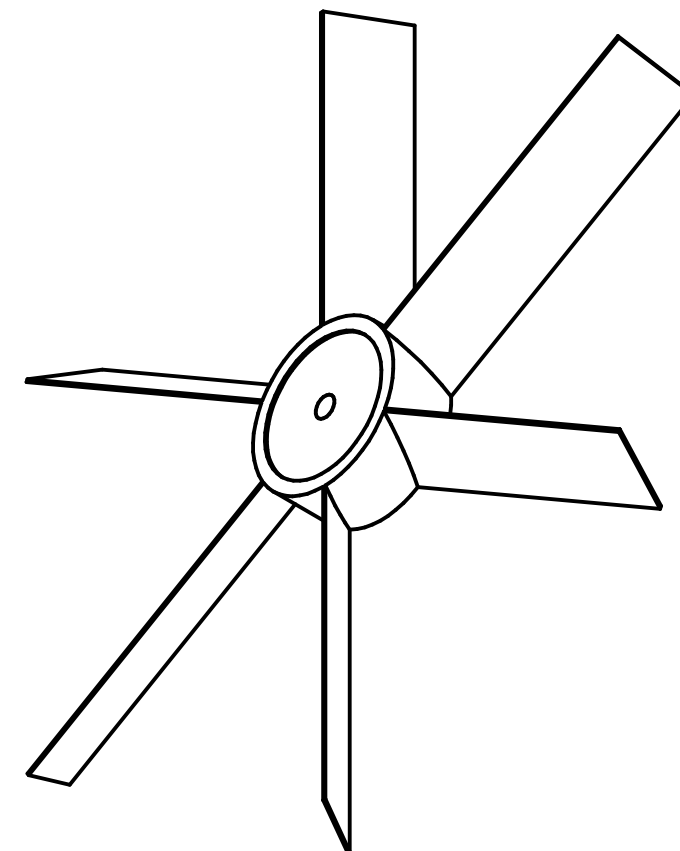
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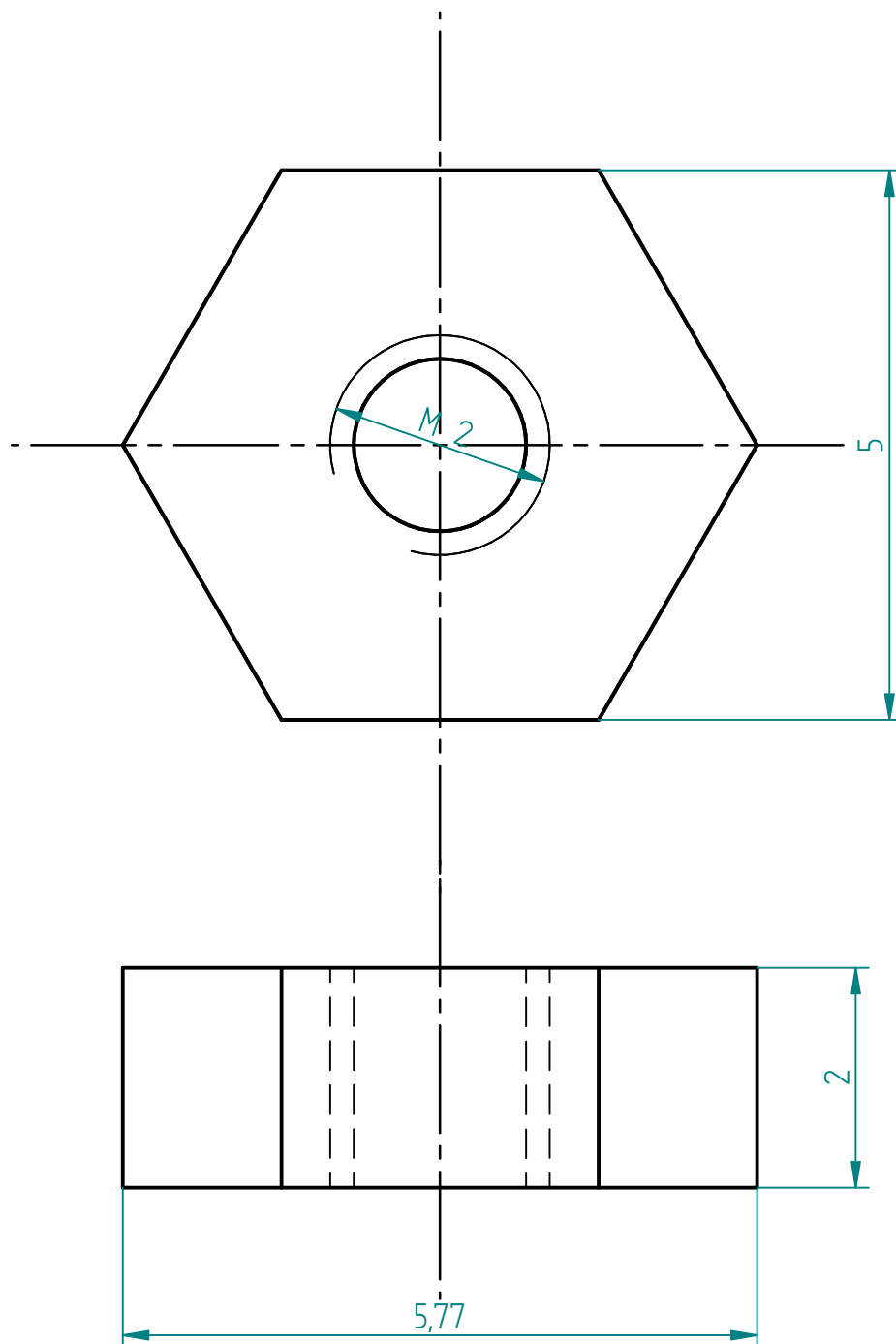
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CORTE A-A



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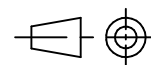
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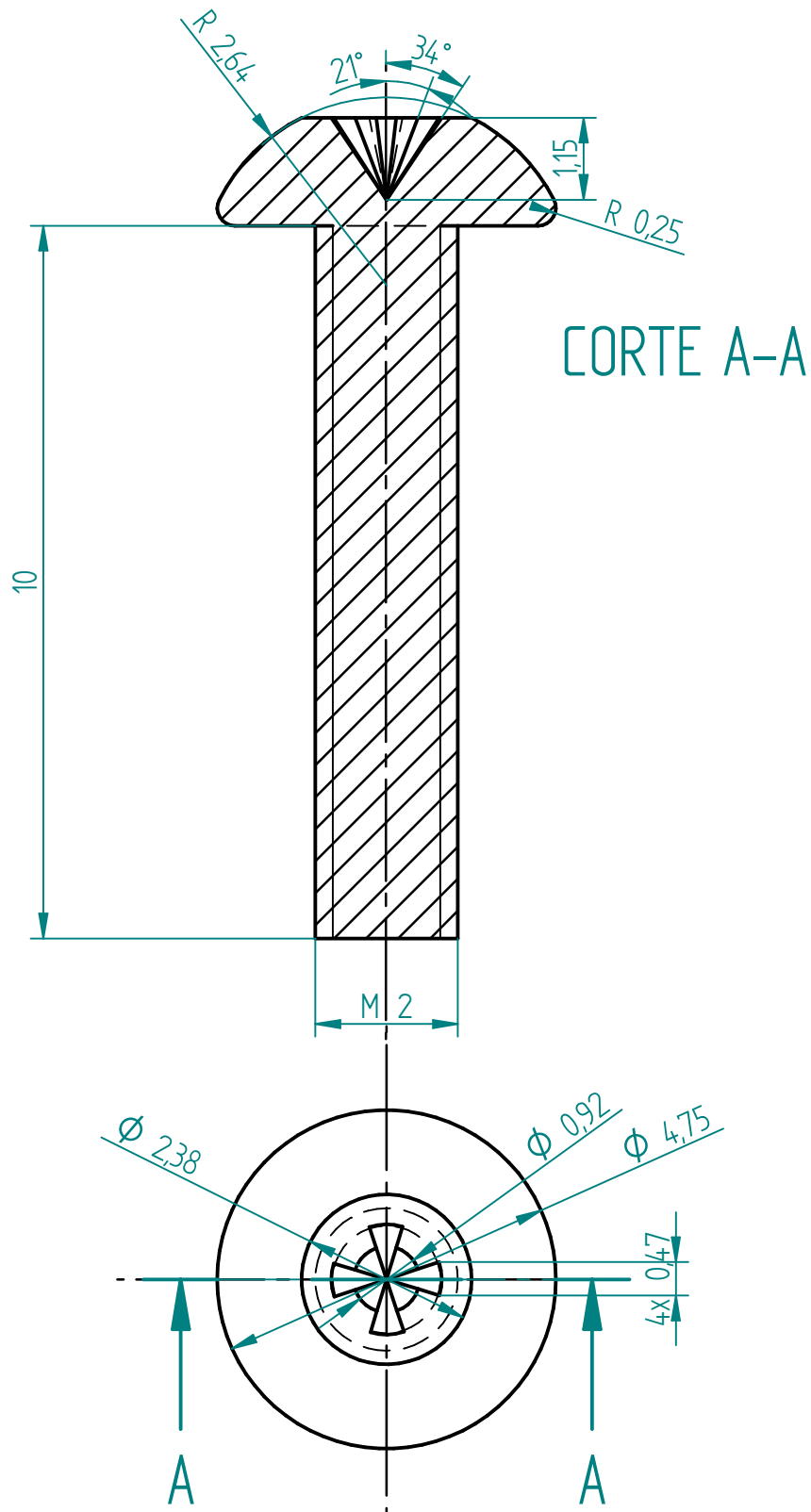
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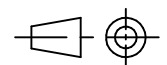
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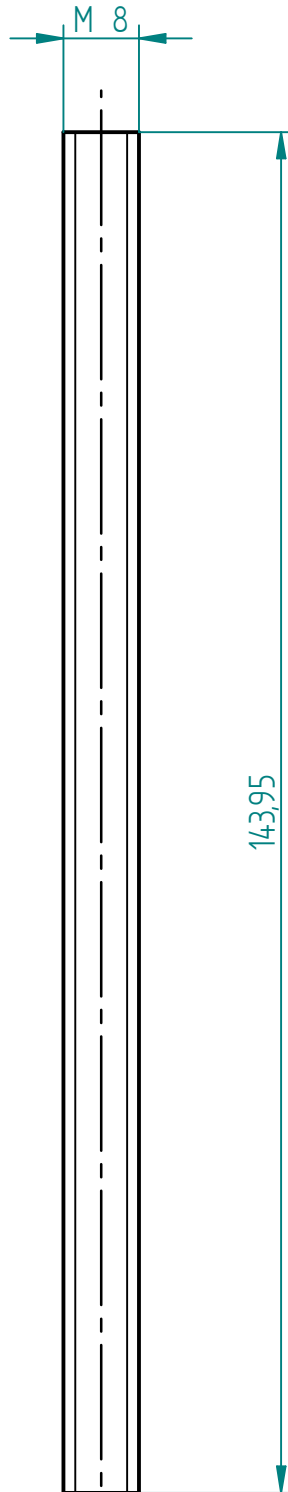
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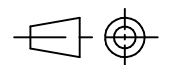
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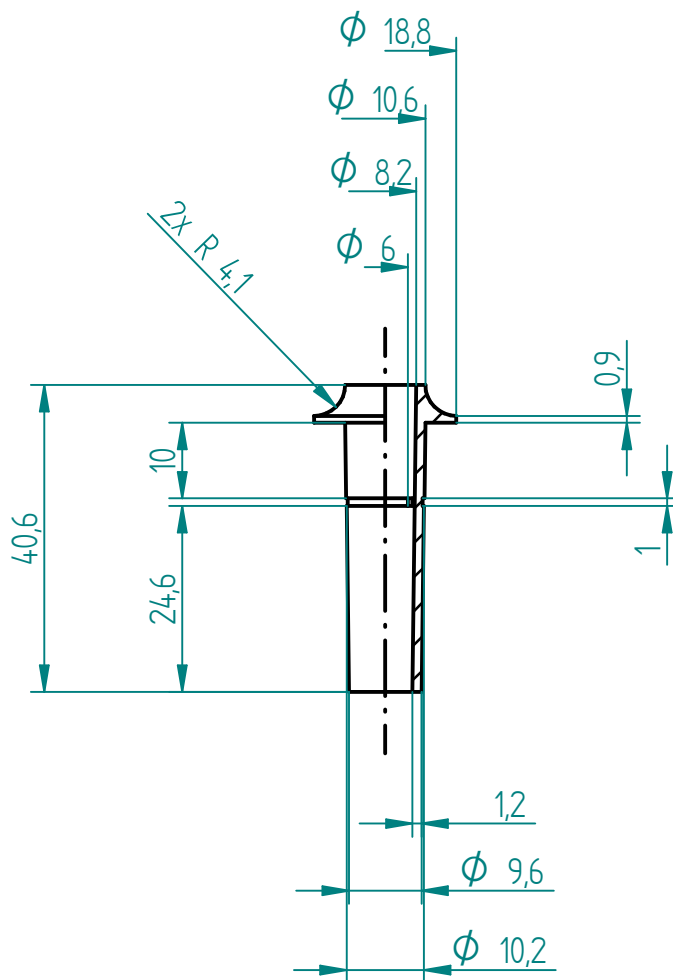
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12/12/2023

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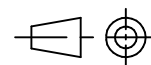


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Title:

button_move_fan



Complementary title:

Scale:

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Created by:

YAJING YE

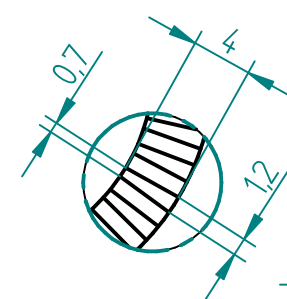
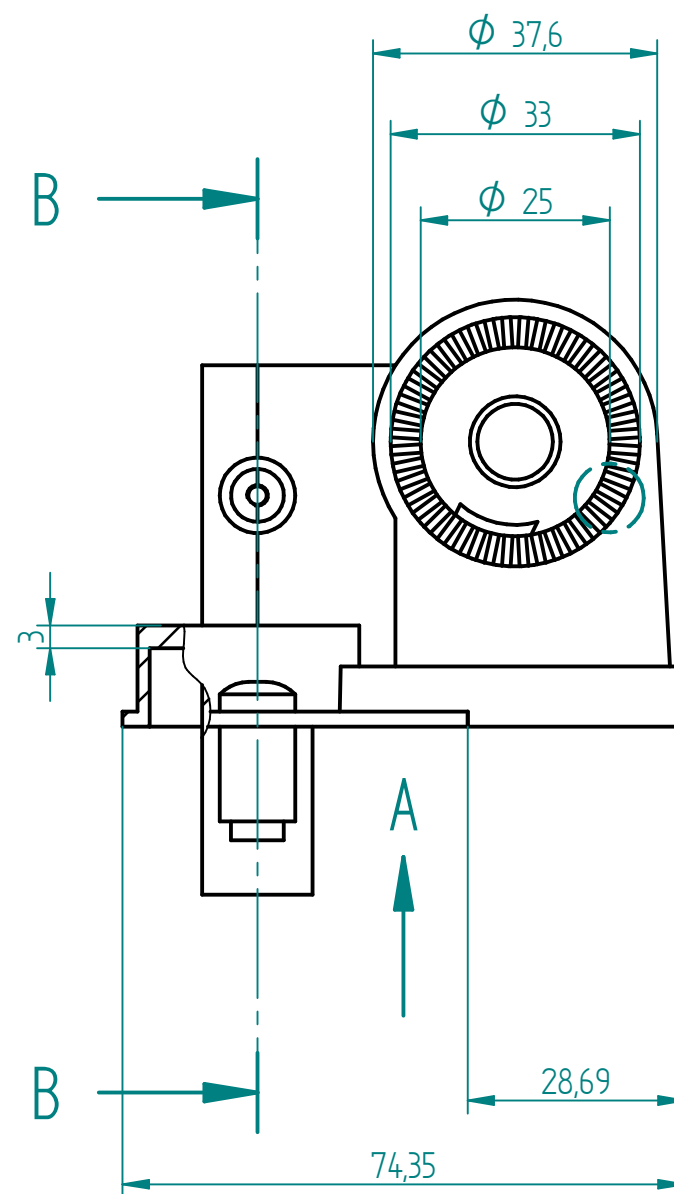
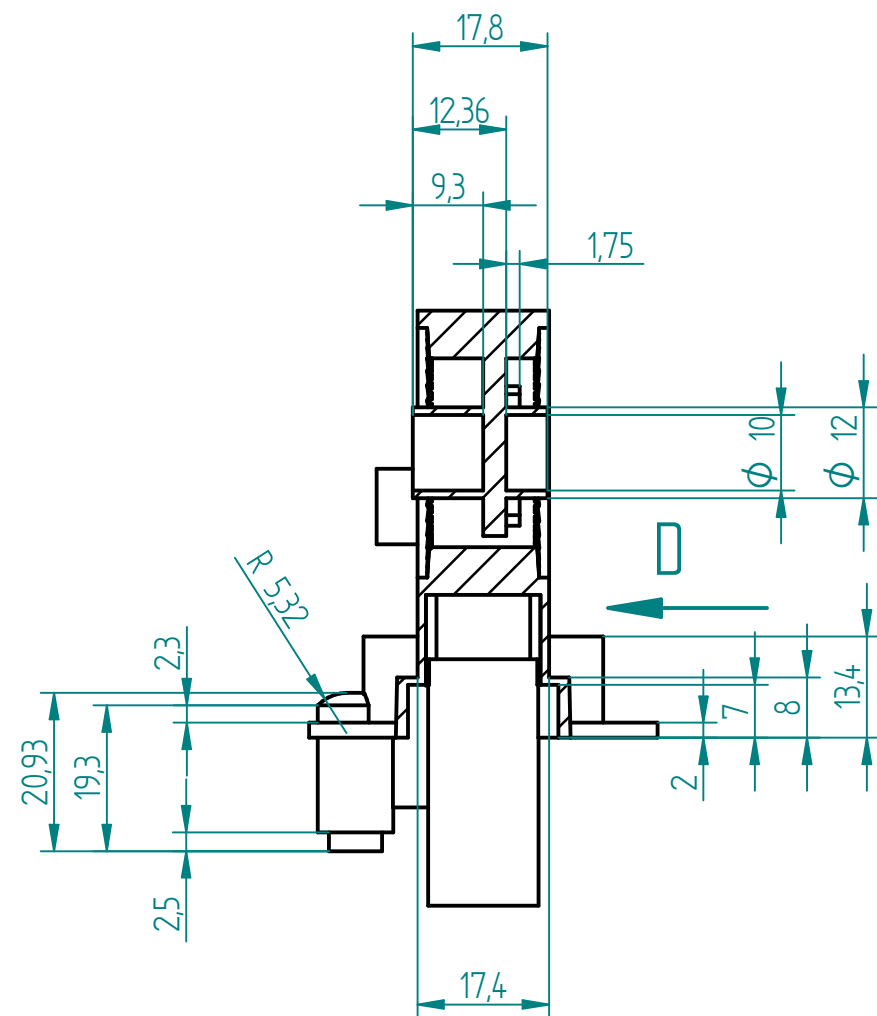
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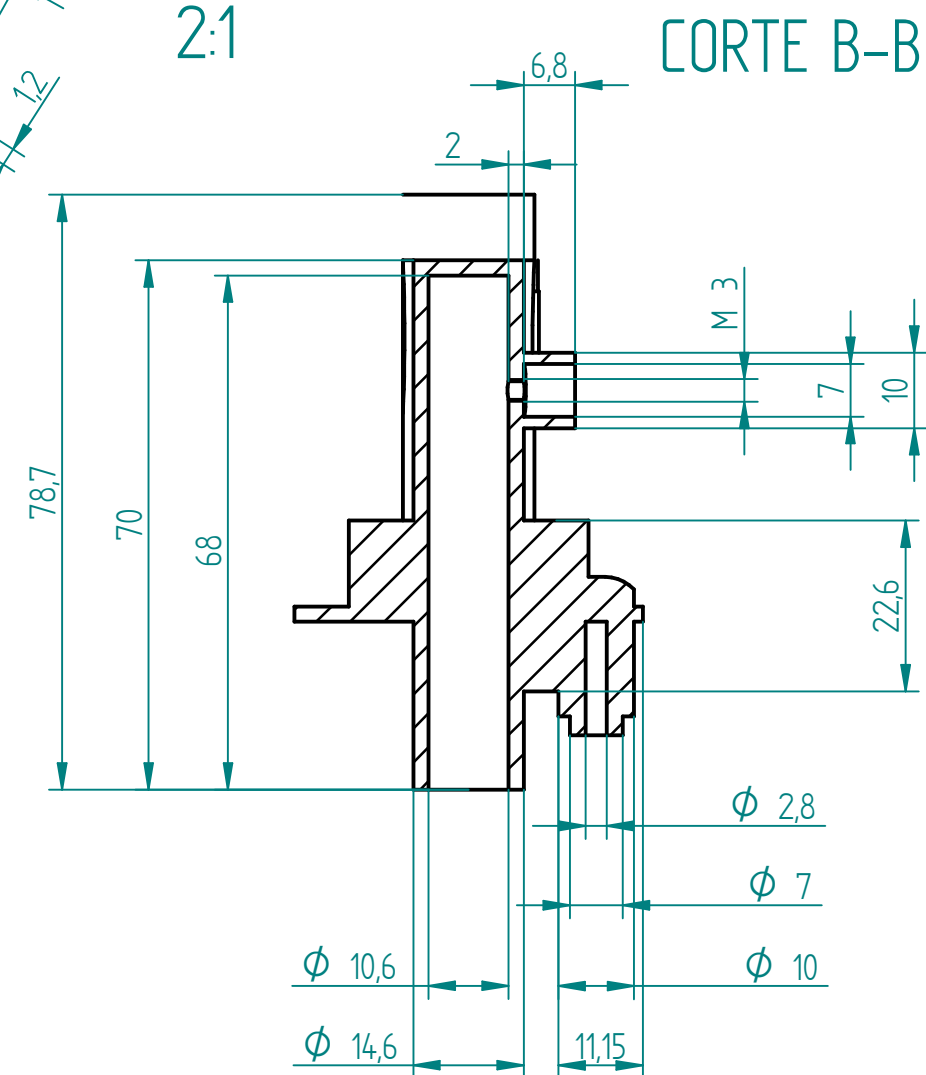
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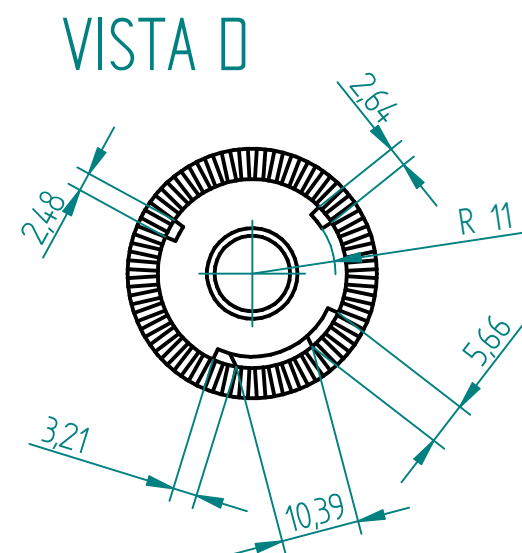
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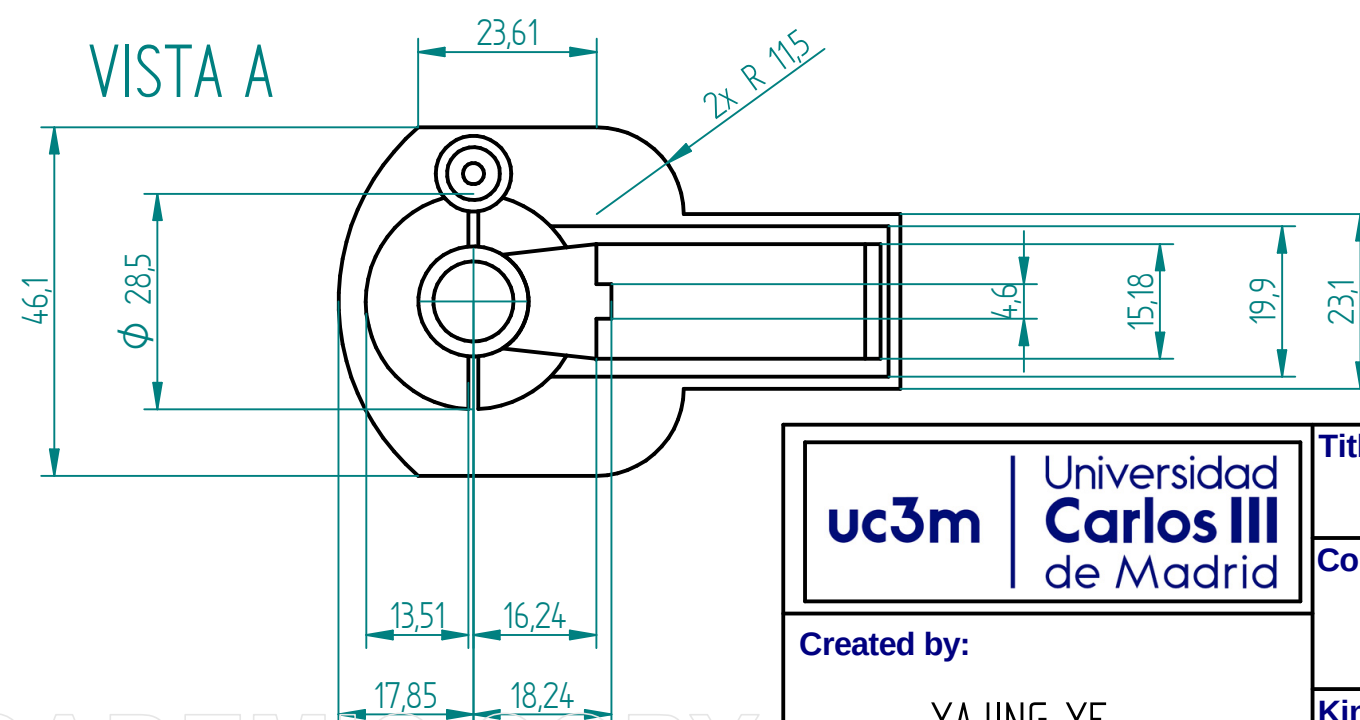
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CORTE B-B



VISTA D



VISTA A

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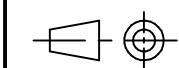
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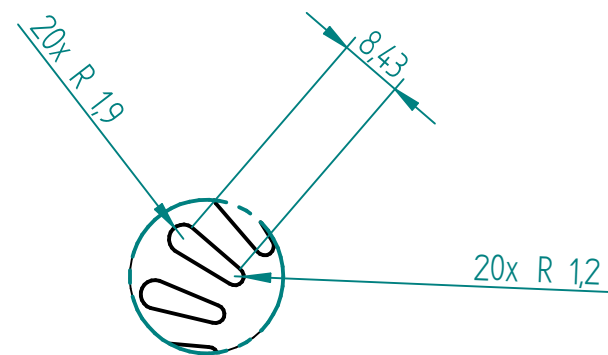
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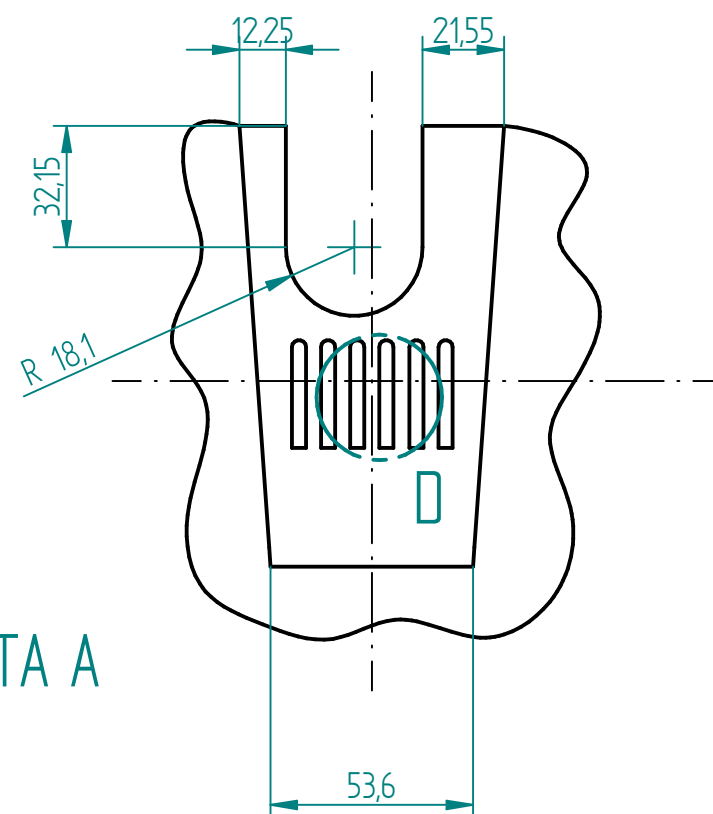
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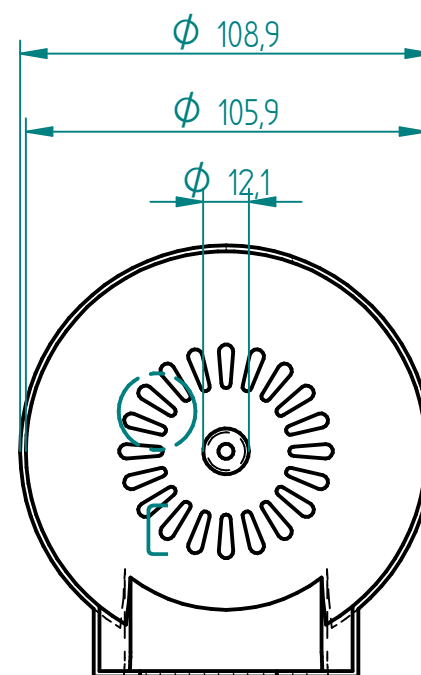
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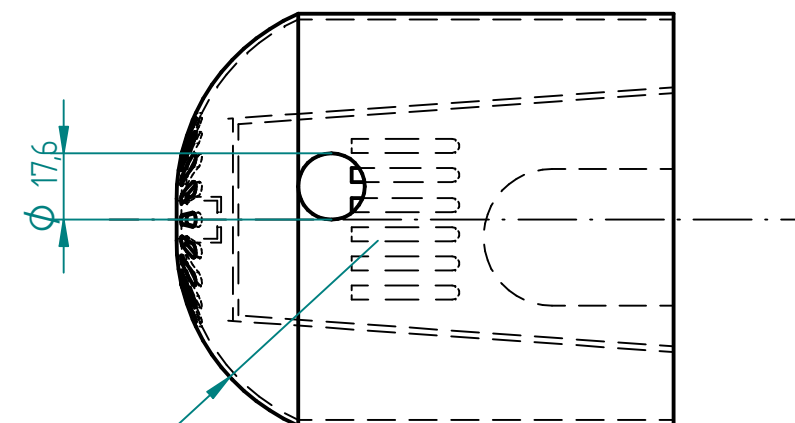
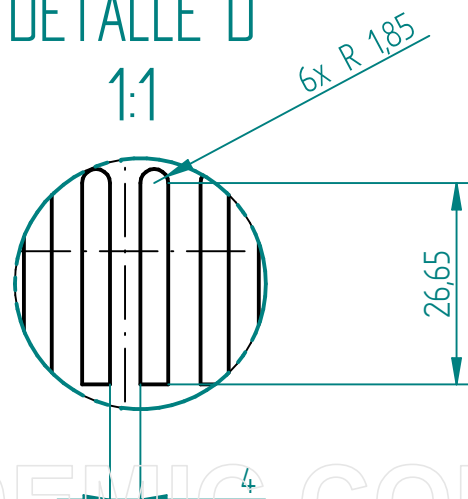


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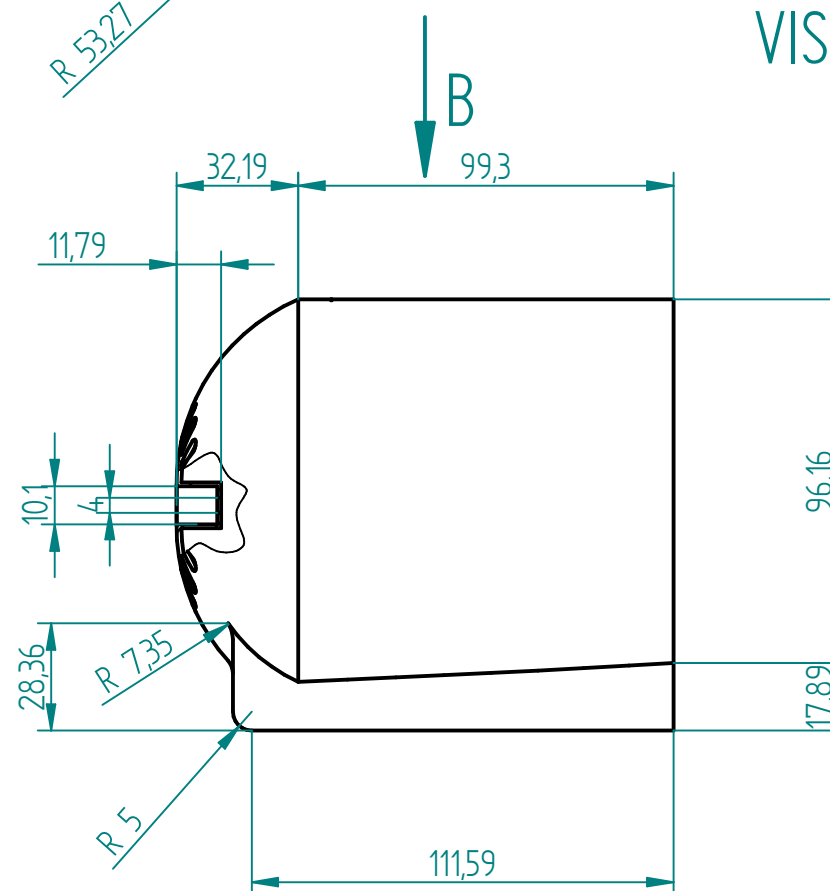


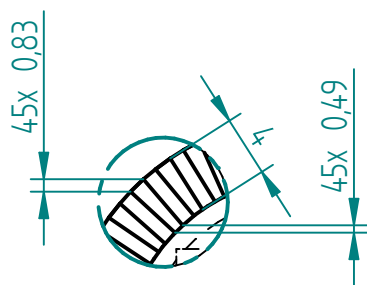
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DETALLE D
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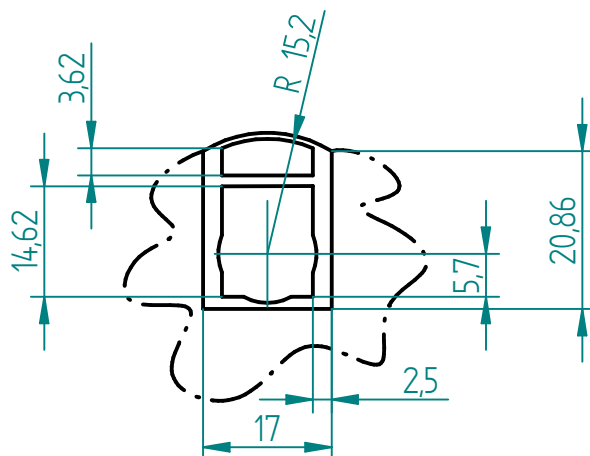
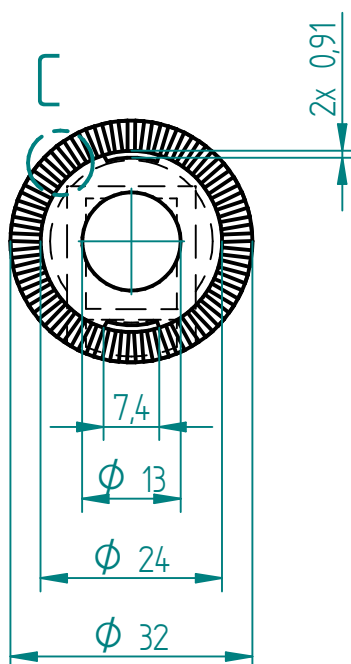


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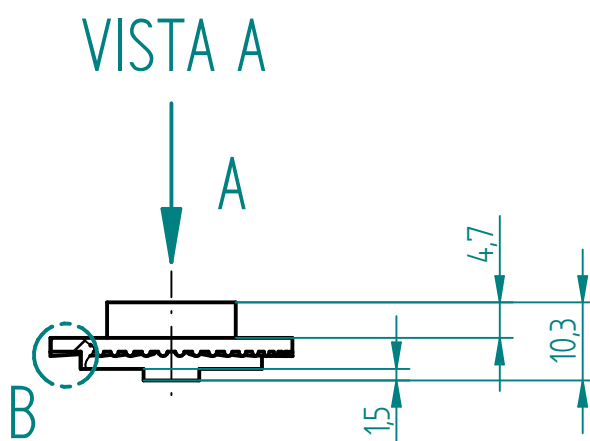
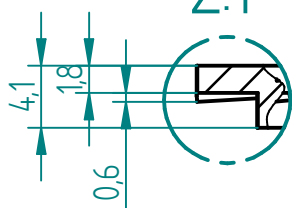




DETALLE C
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DETALLE B
2:1



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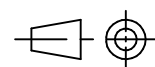
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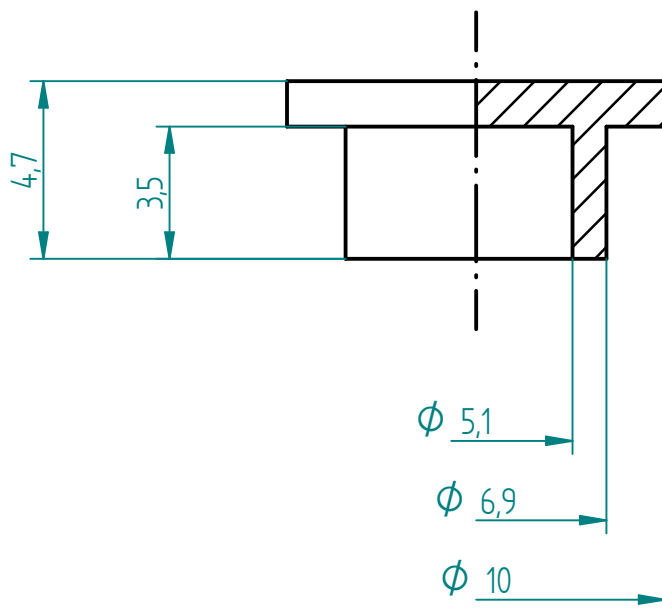
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06/12/2023

Page number:

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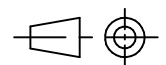
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Title:

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Complementary title:



Scale:

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Created by:

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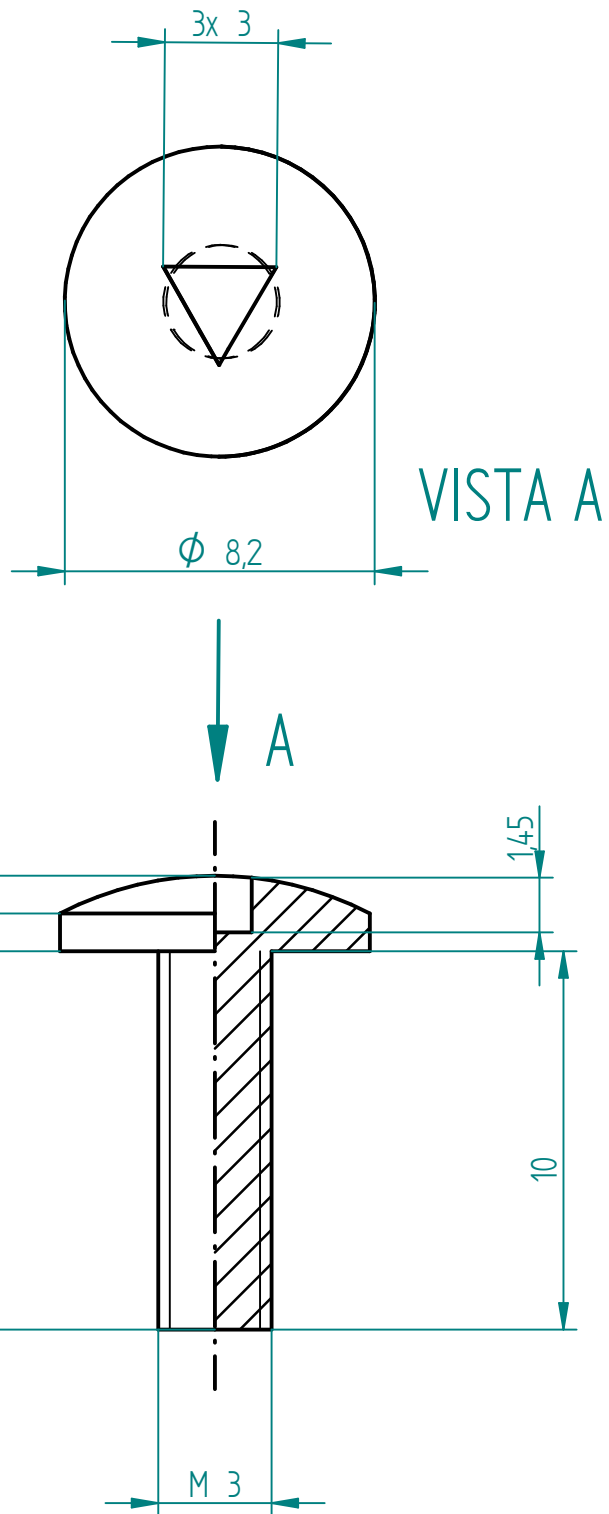
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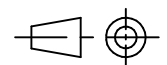
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Title:

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Complementary title:



Scale:

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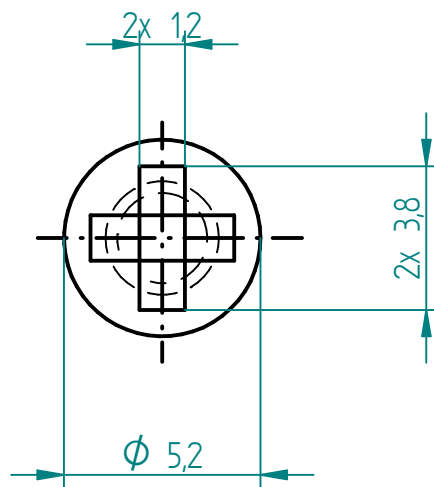
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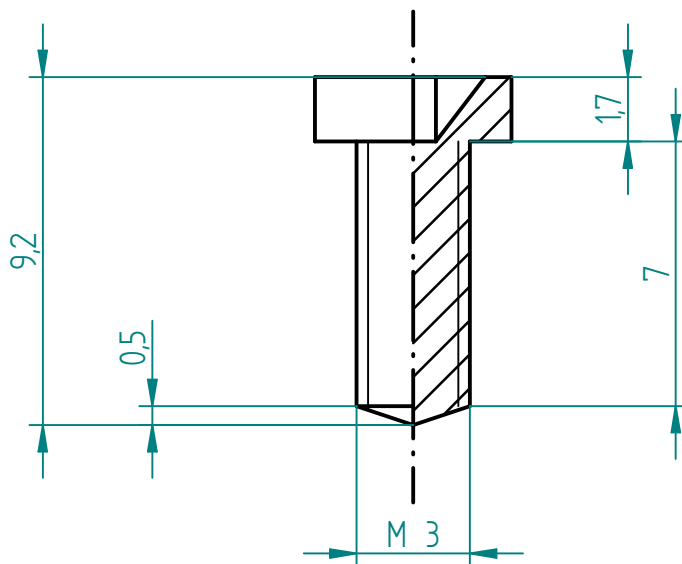
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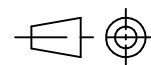
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Title:

screw_piece_move_fan

Complementary title:



Scale:

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06/12/2023

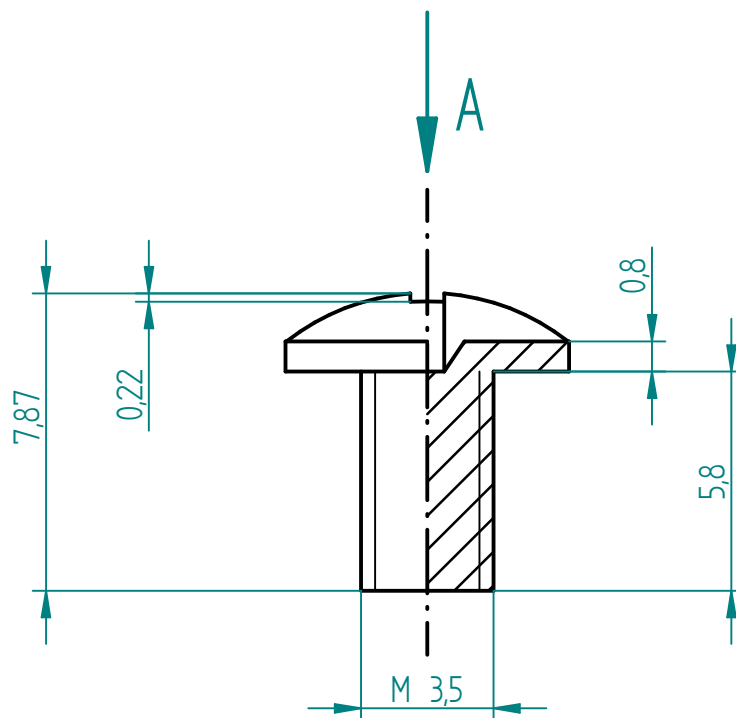
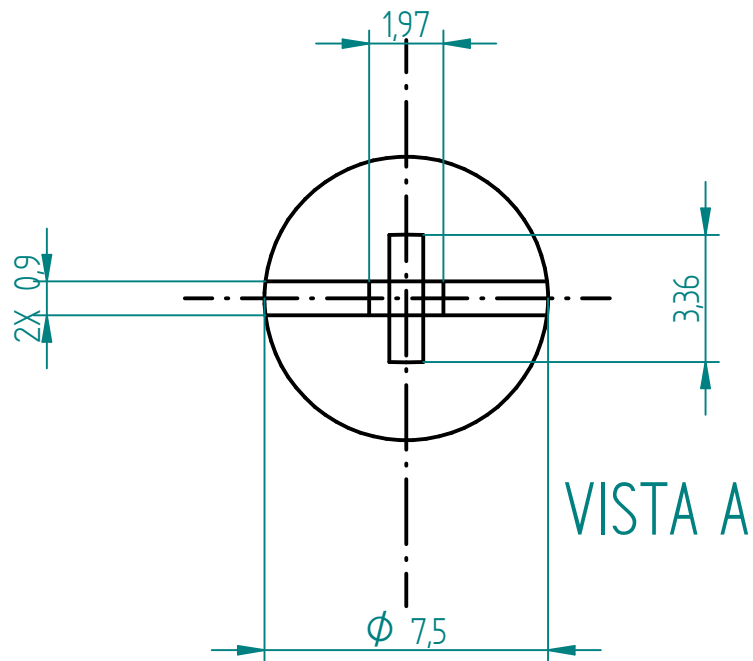
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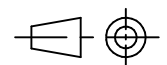
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Title:

screw_motor_top

Complementary title:



Scale:

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Created by:

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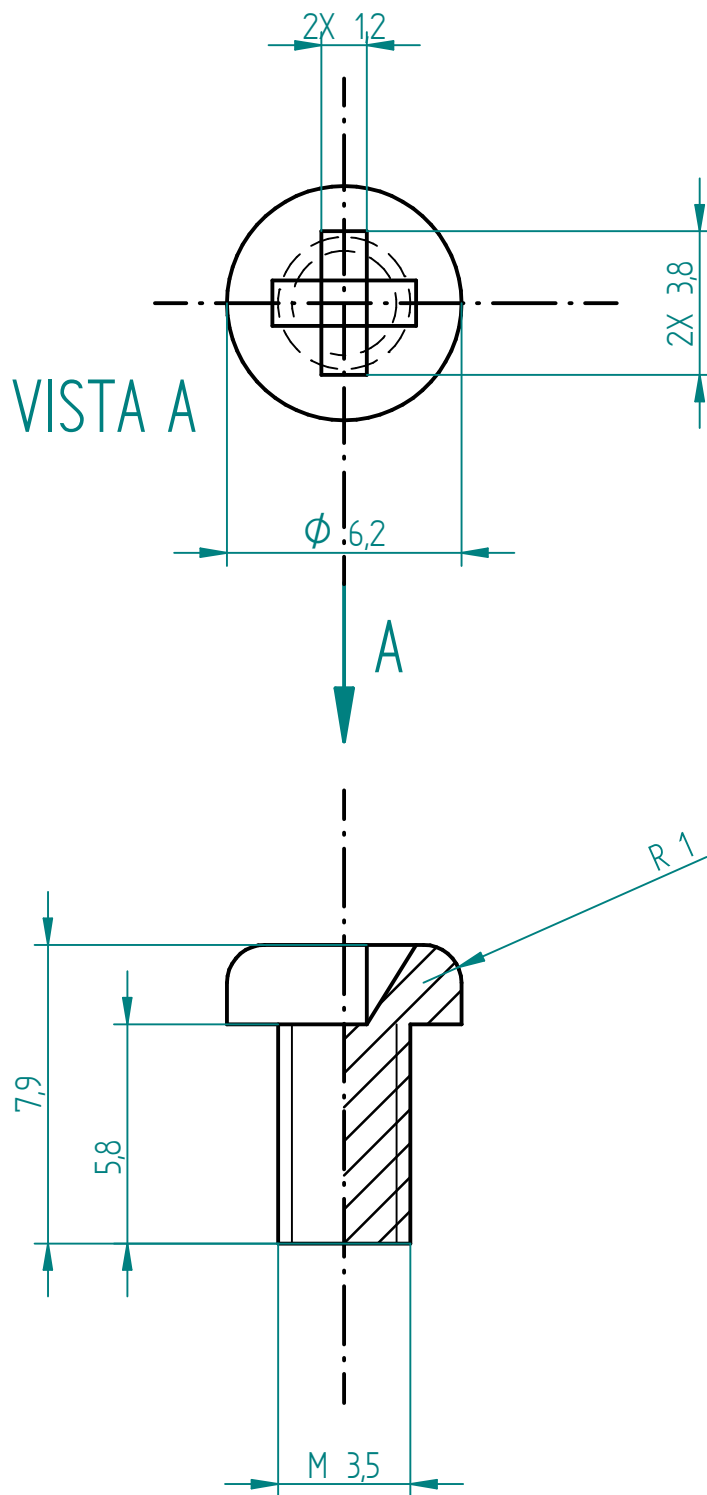
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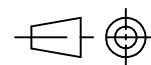
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Title:

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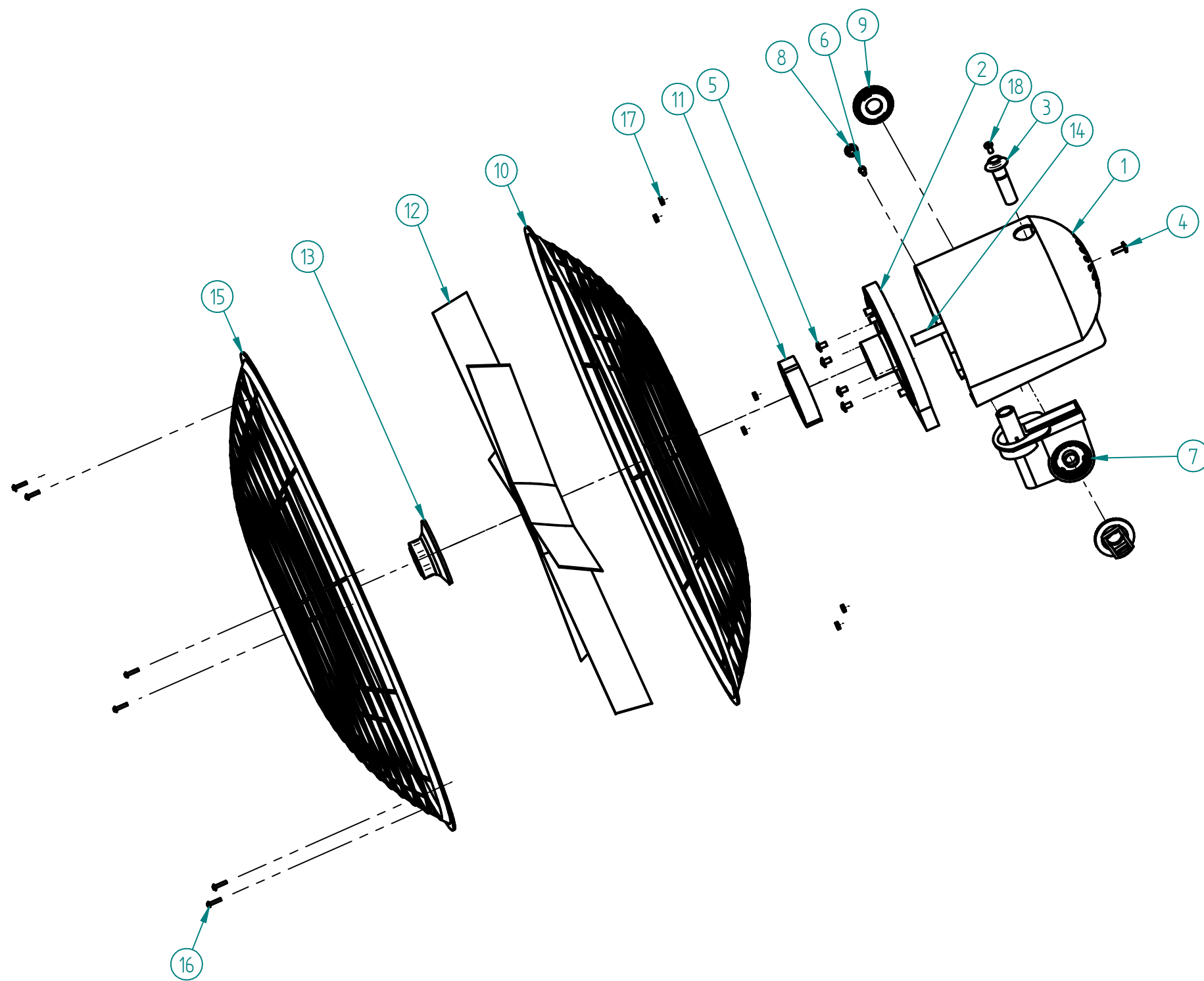
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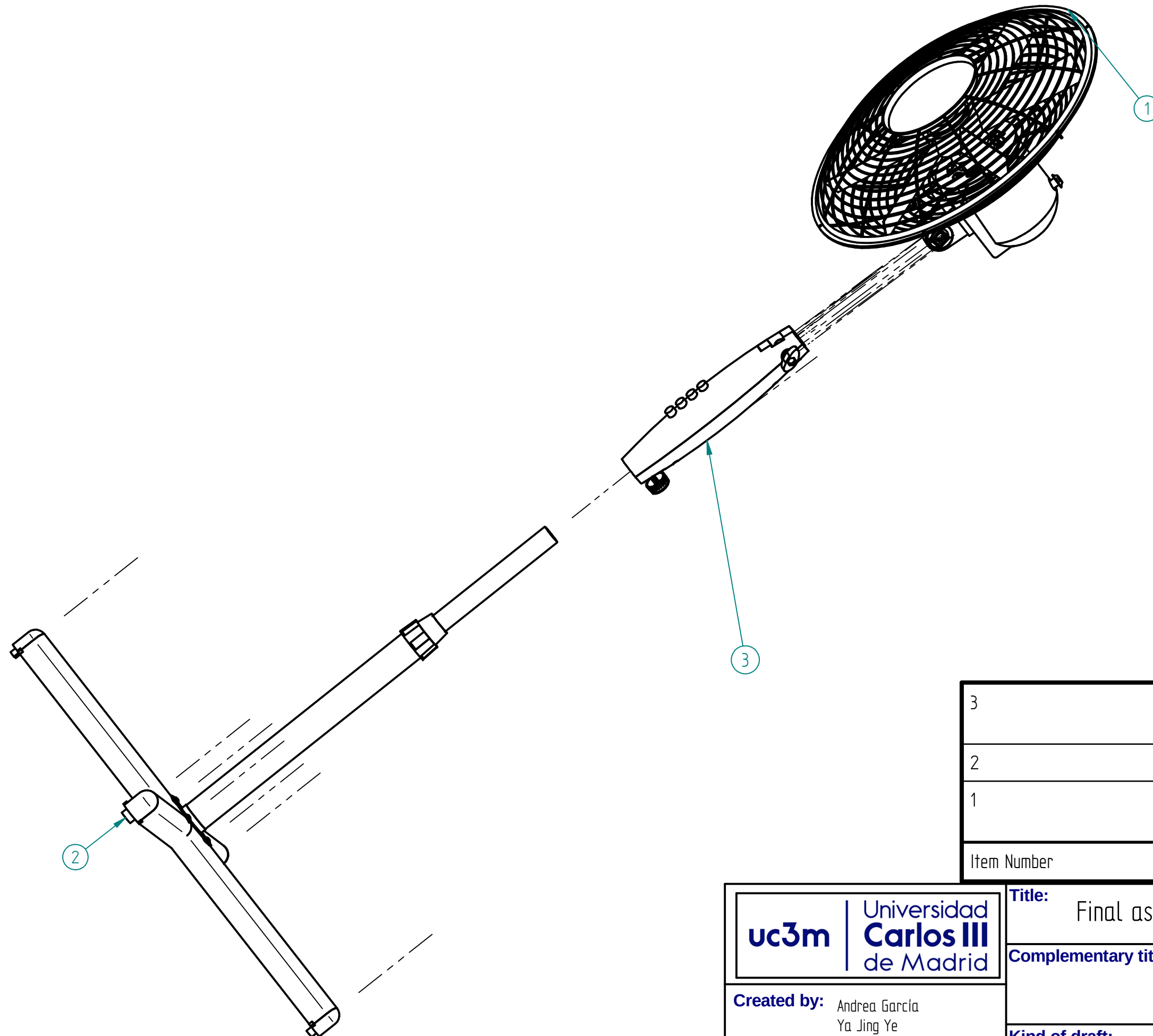
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18	Screw button move fan	1
17	Nut assembly lid	6
16	Screw assembly lid	6
15	Back lid	1
14	Screw assembly helix	1
13	Connector 2	1
12	Helix	1
11	Connector 1	1
10	Front lid	1
9	Round thread	2
8	Top screw piece move fan	1
7	Piece move fan	1
6	Screw piece move fan	1
5	Screw motor top	4
4	Screw motor cover	1
3	Button move fan	1
2	Motor top	1
1	Motor cover	1
Item Number	Title	Cantidad

FINAL ASSEMBLY



3	Central handler subassembly	1
2	Base subassembly	1
1	Helix and motor subassembly	1
Item Number	Title	Cantidad

uc3m Universidad Carlos III de Madrid Created by: Andrea García Ya Jing Ye Lola Martínez Victor Chen	Title: Final assembly	Scale: 1:5 Date: 12-12-23 Page number: 51
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